

CONCISE ALGEBRAIC COMBINATORICS

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1. INTRODUCTION

These are concise notes on Algebraic Combinatorics. The overall structure and content are based primarily on the MIT course 18.212: Algebraic Combinatorics, taught by Alexander Postnikov. These notes are intended to help readers who are familiar with Algebraic Combinatorics recall core ideas, proofs, and the structural flow of the field. Most of the material, with the exception of a few proofs, is drawn directly from the course lectures, and all credit for the original content belongs to the instructor. Detailed proofs are omitted or replaced with proof hints, and for results beyond the scope of these notes, additional references are provided.

2. CATALAN NUMBERS

Definition 2.1 (Dyck path). A *Dyck path of length $2n$* is a sequence $(a_i)_{i=0}^{2n}$ with $a_0 = a_{2n} = 0$, $|a_i - a_{i+1}| = 1$, and $a_i \geq 0$ for all i . Equivalently, it is a lattice path in $\mathbb{Z}^2$ from $(0, 0)$ to $(2n, 0)$ that never goes below the x -axis, where each step $(i, a_i) \rightarrow (i+1, a_{i+1})$ is called an *up-step* U if $a_{i+1} = a_i + 1$ and a *down-step* D if $a_{i+1} = a_i - 1$.

Definition 2.2 (Catalan numbers). The n -th *Catalan number* C_n is the number of Dyck paths of length $2n$.

2.1. Formula for the Catalan Numbers.

Proposition 2.3. *The Catalan numbers satisfy $C_0 = 1$ and*

$$C_n = \sum_{k=1}^n C_{k-1} C_{n-k}.$$

The generating function $f(x) = \sum_{n=0}^{\infty} C_n x^n$ satisfies $f(x) = x f(x)^2 + 1$, with solution

$$f(x) = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

Proof. First-return decomposition of Dyck paths, together with coefficient extraction: $[x^n] f(x)^2 = \sum_{k=0}^n C_k C_{n-k}$. $\square$

Theorem 2.4 (Generalized Binomial Theorem). *For $\alpha \in \mathbb{R}$ and $|y| < 1$,*

$$(1+y)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} y^k, \quad \binom{\alpha}{k} = \frac{\alpha(\alpha-1) \cdots (\alpha-k+1)}{k!}.$$

Lemma 2.5. *The number of non-Dyck paths from $(0, 0)$ to $(2n, 0)$ with n up-steps and n down-steps equals $\binom{2n}{n-1}$.*

Proof. Bijection to paths from $(0, 0)$ to $(2n, -2)$ via reflection method at first touching of $y = -1$; count: $\binom{2n}{n-1}$. $\square$

Definition 2.6 (Dyck⁻ path). Among $\binom{2n+1}{n}$ length- $(2n+1)$ sequences with $(n+1)$ down-steps and n up-steps, a *Dyck⁻ path* has all partial sums ≥ 0 except possibly after the last step (only the final step may go below the x -axis).

Equivalently, a Dyck⁻ path of length $2n + 1$ is a sequence $a_0 = 0, a_1, \dots, a_{2n+1}$ with $|a_i - a_{i+1}| = 1$, $a_i \geq 0$ for $i = 0, \dots, 2n$, and $a_{2n+1} = -1$.

Lemma 2.7 (Cycle Lemma). *Let $a_1, \dots, a_m$ be a sequence of integers summing to 1. Then exactly one of the m cyclic shifts $a_i, a_{i+1}, \dots, a_m, a_1, \dots, a_{i-1}$ has all partial sums positive.*

Proof. Last attainment of the global minimum of the partial sums gives the unique valid starting index; all other shifts fail at that position. $\square$

Corollary 2.8. *Among all $\binom{2n+1}{n}$ sequences of length $2n + 1$ with $(n + 1)$ down-steps and n up-steps, exactly $\frac{1}{2n+1} \binom{2n+1}{n}$ are Dyck⁻ paths. Each equivalence class of cyclic shifts contains exactly one Dyck⁻ path.*

Proof. Encode steps as ± 1 (sum -1). Exactly one cyclic shift of the negated sequence has all partial sums positive, hence exactly one shift of the original is Dyck⁻ (2.7). $\square$

Definition 2.9 (Exceedance). The *exceedance* of a sequence $a_0 = 0, a_1, \dots, a_{2n} = 0$ with n up-steps and n down-steps is the number of up-steps $a_i \rightarrow a_{i+1}$ for which $a_i > 0$, i.e. the number of up-steps taken strictly above the x -axis.

Lemma 2.10. *There is an invertible map on sequences of n up-steps and n down-steps that decreases exceedance by exactly 1. This partitions all $\binom{2n}{n}$ sequences into $n + 1$ equally sized classes (exceedances $0, 1, \dots, n$), giving*

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

Proof. First return after the first exceedance; swap the portion of the path; recoverable by finding the last down-step from the x -axis. $\square$

Theorem 2.11. $C_n = \frac{1}{n+1} \binom{2n}{n}$.

Proof. Any of the following: 2.3 + 2.4. 2.5. 2.8. 2.10. $\square$

2.2. Many Bijections.

Theorem 2.12. *The number of complete parenthesizations of a product of $n + 1$ variables is C_n .*

Proof. First-return decomposition of Dyck paths; left sub-path $\leftrightarrow$ left factor, right sub-path $\leftrightarrow$ right factor. $\square$

Definition 2.13 (Binary tree). A *binary tree* on n vertices is a tree where each vertex has at most two ordered (left/right) children; internal nodes need not have both children. The *leaves* are nodes with no children.

Definition 2.14 (Full binary tree). A *full binary tree* is a binary tree in which every internal node has exactly two children.

Theorem 2.15. *The number of full binary trees with $n + 1$ leaves is C_n .*

Proof. Bijection with parenthesizations (2.12): label leaves left-to-right $x_0, \dots, x_n$; each internal node corresponds to a binary operation. Alternatively, DFS over the tree: up-step when has child, down-step when is leaf. $\square$

Theorem 2.16. *The number of triangulations of a convex $(n + 2)$ -gon is C_n .*

Proof. Bijection with full binary trees: fix a root edge; $n + 1$ remaining boundary edges $\leftrightarrow$ leaves; n triangles $\leftrightarrow$ internal nodes. $\square$

Theorem 2.17. *The number of non-crossing perfect matchings on $2n$ labeled points in convex position is C_n .*

Proof. Walk the vertices in order; unmatched partner $\rightarrow$ up-step U , matched partner already seen $\rightarrow$ down-step D ; non-crossing $\Leftrightarrow$ Dyck path. $\square$

Theorem 2.18. *The number of non-nesting perfect matchings on $2n$ labeled points in convex position is C_n .*

Proof. Bijection with non-crossing matchings (2.17): scan left to right, exchanging crossings and nestings. $\square$

Theorem 2.19. *The number of non-crossing set partitions of n labeled points in convex position is C_n .*

Proof. Bijection with non-crossing matchings (2.17): expand each vertex v_i to a pair (v_i^c, v_i^o) ; connect $v_{i,j}^o$ to $v_{i,j+1}^c$ cyclically within each block. $\square$

Theorem 2.20. *The number of binary trees on n vertices is C_n .*

Proof. Bijection with full binary trees: add two children to each leaf, one child to each unary node $\Rightarrow$ full binary tree with $n + 1$ leaves. Alternatively, recursive Dyck path decomposition: last return to origin splits path into left and right subtree. $\square$

Definition 2.21 (σ -avoidance). For $w \in S_n$ and $\sigma \in S_m$ with $m \leq n$, say w is σ -avoiding if there is no subsequence of indices $i_1 < \dots < i_m$ such that $w(i_1), \dots, w(i_m)$ have same order as $\sigma(1), \dots, \sigma(m)$.

Definition 2.22 (Stack-sortable). A permutation $w \in S_n$ is *stack-sortable* if it can be sorted to the identity by a sequence of push/pop operations on a single LIFO stack reading w left to right.

Theorem 2.23. *A permutation $w \in S_n$ is stack-sortable if and only if it is 231-avoiding. In particular, the number of stack-sortable permutations in S_n is C_n .*

Proof. Greedy algorithm: pop all elements smaller than the next input before pushing; 231 pattern forces incorrect output order; failed sort $\Rightarrow$ 231 in w . Bijection with Dyck paths: operations to steps. $\square$

Definition 2.24 (k -queue-sortable). A permutation $w \in S_n$ is *k -queue-sortable* if it can be sorted to the identity by k consecutive passes through a FIFO queue. One may enqueue the current element or dequeue the front to output at each step.

Lemma 2.25. *A permutation $w \in S_n$ is k -queue-sortable if and only if w is $(k + 1, k, \dots, 1)$ -avoiding.*

Proof. Greedy algorithm: enqueue the current input to the first queue whose last element is smaller than the input; pop elements from the front in order; $(k + 1, k, \dots, 1)$ pattern forces incorrect output order; failed sort $\Rightarrow$ at the moment of failure the last queue elements compose $(k + 1, k, \dots, 1)$ in w . $\square$

Theorem 2.26. *The number of 2-queue-sortable permutations in S_n is C_n .*

Proof. Count number of 321-avoiding permutations (2.25); it is equal to the number of 123-avoiding permutations (3.22). $\square$

Theorem 2.27. $\forall \sigma \in S_3$, *the number of σ -avoiding permutations in S_n is C_n .*

Proof. Reverse and complement $w(i) \mapsto n + 1 - w(i)$ reduce to the stack-sortable or 2-queue-sortable cases (2.23, 2.25, 2.26). $\square$

2.3. Narayana Numbers.

Definition 2.28 (Peak). A *peak* of a Dyck path is a consecutive pair of steps UD (an up-step immediately followed by a down-step).

Corollary 2.29. $\sum_{k=1}^n N(n, k) = C_n$.

Proof. Each Dyck path of length $2n$ has a peak, so peak count partitions the C_n paths. $\square$

Definition 2.30 (Narayana numbers). The *Narayana number* $N(n, k)$ is the number of Dyck paths of length $2n$ with exactly k peaks. They are displayed in the *Narayana triangle*:

				1						
				1		1				
			1		3		1			
		1		6		6		1		
	1		10		20		10		1	
1		15		50		50		15		1

Theorem 2.31. $N(n, k)$ equals the number of binary trees on n vertices with exactly $k - 1$ left edges.

Proof. Via binary trees $\leftrightarrow$ full binary trees with $n + 1$ leaves and right-first DFS $\leftrightarrow$ Dyck paths, peaks count right leaves, hence left edges in the original tree. $\square$

Corollary 2.32. $N(n, k) = N(n, n + 1 - k)$.

Proof. Flip the tree: left edges become right edges. $\square$

Theorem 2.33 (Narayana Formula).

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}.$$

Proof. Consider sequences of n up-steps and $n + 1$ down-steps with k peaks. Their count is $\frac{1}{n} \binom{n-1}{k-1} \binom{n}{k}$. By shifting UD groups, $\frac{1}{k}$ of them are Dyck paths (2.7). $\square$

3. YOUNG TABLEAUX

Definition 3.1 (Partition). A *partition* of a nonnegative integer n is a sequence

$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$$

of positive integers such that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0, \quad \lambda_1 + \lambda_2 + \dots + \lambda_k = n.$$

We write $\lambda \vdash n$. The quantity $|\lambda| = n$ is called the *size* of the partition.

Definition 3.2 (Young diagram). A *Young diagram* of size n is a way to identify a partition λ with size n by the set of cells $\{(i, j) : 1 \leq i \leq k, 1 \leq j \leq \lambda_i\}$.

Definition 3.3 (Standard Young tableau). A *standard Young tableau* (SYT) of shape $\lambda \vdash n$ is a bijection $T : \lambda \rightarrow \{1, \dots, n\}$ that is strictly increasing along every row (left to right) and every column (top to bottom). Equivalently, it is a chain $\emptyset = \lambda^{(0)} \subset \lambda^{(1)} \subset \dots \subset \lambda^{(n)} = \lambda$ of Young diagrams in which each $\lambda^{(i)} \setminus \lambda^{(i-1)}$ consists of exactly one cell; the label of that cell is i .

Proposition 3.4. *The number of SYTs of shape $n \times 2$ is C_n .*

Proof. Bijection with Dyck paths: the entries of the first row (in increasing order) are the indices of the up-steps; the entries of the second row are the indices of the down-steps. $\square$

3.1. Hook Length Formula.

Definition 3.5 (Hook). For a cell $b = (i, j) \in \lambda$, the *hook* of b is

$$H(b) = \{(i, j') \in \lambda : j' \geq j\} \cup \{(i', j) \in \lambda : i' \geq i\},$$

and the *hook length* $h(b) = |H(b)|$.

Definition 3.6 (Hook walk). A *hook walk* on λ starting from the top-left cell $b_1 = (1, 1)$ is a random walk $b_1, b_2, \dots$ where at each step from b_k one chooses uniformly at random among the $h(b_k) - 1$ cells in $H(b_k) \setminus \{b_k\}$, terminating when a *corner* of λ is reached (a cell with no cell below or to its right).

Definition 3.7 (Cohook). For a corner c of λ , the *cohook* of c is the minimal rectangle in λ containing $(1, 1)$ and c . Label the cells along the bottom edge of the cohook $X_1, \dots, X_k$ (left to right) and those along the right edge $Y_1, \dots, Y_l$ (top to bottom), and set $x_i = h(X_i) - 1$ and $y_j = h(Y_j) - 1$. For any cell (i, j) in the cohook, $h(i, j) - 1 = x_i + y_j$, since for any rectangle $\{b, x, y, c\}$ with b top-left and c bottom-right, $h(b) + h(c) = h(x) + h(y)$.

Definition 3.8 (Lattice walk). A *lattice walk* over a grid S is a path from the top-left cell of S to bottom-right cell using only right-steps and down-steps.

Lemma 3.9. *For a $k \times l$ rectangle with bottom-edge hook lengths $x_1, \dots, x_k$ and right-edge hook lengths $y_1, \dots, y_l$,*

$$\sum_{\text{lattice walks } (i, j) \text{ in walk}} \prod \frac{1}{x_i + y_j} = \prod_{i=1}^k \frac{1}{x_i} \cdot \prod_{j=1}^l \frac{1}{y_j}.$$

Proof. Induction on $k + l$. Base case $k = 1$: the unique walk gives $\prod_{i=1}^n \frac{1}{x_i}$. Inductive step: any walk passes through either the (k, l) cell via the $(k - 1) \times l$ or the $k \times (l - 1)$ sub-rectangle. $\square$

Lemma 3.10. *The probability that the hook walk on λ terminates at corner c is*

$$\frac{\prod_{b \in \lambda} h(b)}{n \cdot \prod_{b \in \lambda \setminus c} h(b)}.$$

Proof. Decompose by summing over all rectangular subgrids of the cohook of c .

$$\sum_{\text{hook walks to } c \text{ in hook walk}} \prod \frac{1}{h(b) - 1} = \prod_{i=1}^k \left(1 + \frac{1}{x_i}\right) \prod_{j=1}^l \left(1 + \frac{1}{y_j}\right) = \frac{\prod_{b \in \lambda} h(b)}{n \cdot \prod_{b \in \lambda \setminus c} h(b)},$$

where the last equality uses that deleting c drops $h(X_i)$ and $h(Y_j)$ by 1 each and fixes the remaining hooks. $\square$

Theorem 3.11 (Hook Length Formula). *For $\lambda \vdash n$, let f^λ denote the number of SYTs of shape λ . Then*

$$f^\lambda = \frac{n!}{\prod_{b \in \lambda} h(b)}.$$

Proof. Both f^λ and $\frac{n!}{\prod_{b \in \lambda} h(b)}$ equal 1 for $n = 1$; recurrence $f^\lambda = \sum_{c \text{ corner}} f^{\lambda \setminus c}$ also holds for $\frac{n!}{\prod_{b \in \lambda} h(b)}$ (3.10). $\square$

3.2. RSK Correspondence.

Construction 3.12 (Schensted insertion algorithm). Given $w \in S_n$, process permutation w in order to build same-shape SYTs P (*insertion*) and Q (*recording*). Their final common shape is $\lambda(w)$, the *Schensted shape*.

P: In row 1, replace the leftmost $y > x$ by x and bump y to row 2, etc.; if none, append x to row 1. Exactly one new cell appears.

Q: When $w(i)$ creates cell (r, c) in P , put i at (r, c) in Q .

Theorem 3.13 (Robinson–Schensted–Knuth Correspondence). *3.12 defines a bijection RSK between S_n and pairs (P, Q) of same-shape SYTs with n cells, i.e. pairs where $P, Q \in \text{SYT}(\lambda)$ for some $\lambda \vdash n$.*

Proof. Valid SYT pair; recover w by reverse bumping using Q for deletion order. $\square$

Lemma 3.14. *Let B_i be the set of values placed into column i of row 1 during 3.12. The elements of B_i appear in strictly decreasing order in w .*

Proof. P -tableau construction from 3.12. $\square$

Lemma 3.15. *For each $x \in B_i$ with $i > 1$, there exists $y \in B_{i-1}$ with $y < x$ appearing before x in w .*

Proof. Let y be row 1, column $i - 1$ when x enters column i ; then $y \in B_{i-1}$, $y < x$, and y precedes x in w . $\square$

Proposition 3.16. *The length λ_1 of the first row of $\lambda(w)$ equals the length of the longest increasing subsequence of w .*

Proof. Any increasing subsequence of length $> \lambda_1$ contains two elements from the same B_i (3.14). Taking $x_{\lambda_1} \in B_{\lambda_1}$ and chaining back gives an increasing subsequence of length λ_1 (3.15). $\square$

Definition 3.17 (Conjugate partition). The *conjugate* λ' has $\lambda'_j = |\{i : \lambda_i \geq j\}|$ (column lengths), i.e. transpose the diagram of λ .

Lemma 3.18. *If $\text{RSK}(w) = (P, Q)$ and w^r denotes the reverse of w , then*

$$P(w^r) = P(w)^T.$$

In particular, $\lambda(w^r) = \lambda(w)'$.

Proof. Reversal swaps row and column insertion; column insertion into P equals row insertion into P^T followed by transpose, so inductively $P(w^r) = P(w)^T$. $\square$

Proposition 3.19. *The length λ'_1 of the first column of $\lambda(w)$ equals the length of the longest decreasing subsequence of w .*

Proof. Decreasing subsequence of w of length k is an increasing subsequence of w^r of length k ; $\lambda_1(w^r) = \lambda'_1(w)$ (3.18, 3.16). $\square$

Proposition 3.20. *If $\text{RSK}(w) = (P, Q)$ then $\text{RSK}(w^{-1}) = (Q, P)$.*

Proof. See [8]. $\square$

Corollary 3.21. $\sum_{\lambda \vdash n} (f^\lambda)^2 = n!$

Proof. RSK is a bijection between S_n and pairs $(P, Q) \in \text{SYT}(\lambda) \times \text{SYT}(\lambda)$ over all $\lambda \vdash n$ (3.13); the left-hand side counts such pairs. $\square$

Corollary 3.22. *The number of 123-avoiding permutations in S_n is C_n .*

Proof. 123-avoiding $\Leftrightarrow \lambda_1(w) \leq 2$ (3.16). For $\text{RSK}(w) = (P, Q)$, keep P , replace entries v of Q by $2n+1-v$, rotate Q 180° below P to get an $n \times 2$ SYT; count C_n (3.4). $\square$

Theorem 3.23 (Greene's Theorem). *Let $\lambda(w) = (\lambda_1, \lambda_2, \dots)$ be the Schensted shape of $w \in S_n$. For any $k \geq 1$,*

$$\lambda_1 + \dots + \lambda_k = \max\{|A_1 \cup \dots \cup A_k| : A_1, \dots, A_k \text{ are increasing subsequences of } w\}.$$

Similarly, if $\lambda' = (\lambda'_1, \lambda'_2, \dots)$ is the conjugate shape, then $\lambda'_1 + \dots + \lambda'_k$ equals the maximum size of a union of k decreasing subsequences of w .

Proof. See [5]. $\square$

4. STATISTICS ON PERMUTATIONS

Notation 4.1 (Permutation notations). Let $w \in S_n$.

- *One-line notation:* $w = w(1), w(2), \dots, w(n)$.
- *Two-line notation:* $w = \begin{pmatrix} 1 & 2 & \dots & n \\ w(1) & w(2) & \dots & w(n) \end{pmatrix}$.
- *Cycle notation:* write w as a product of disjoint cycles $(a_1 a_2 \dots a_k)$ where $w(a_i) = a_{i+1}$ and $w(a_k) = a_1$.

Definition 4.2 (Statistic). A *statistic* on S_n is a map $\sigma : S_n \rightarrow \mathbb{Z}_{\geq 0}$.

4.1. Eulerian Numbers.

Definition 4.3 (Eulerian numbers). For $n, k \geq 0$, the *Eulerian numbers* $A(n, k)$ are defined by the generating function identity

$$f_n(x) = \sum_{j=0}^{\infty} (j+1)^n x^j = \frac{A_n(x)}{(1-x)^{n+1}},$$

where $A_n(x) = \sum_{k=0}^{n-1} A(n, k)x^k$ is the n -th *Eulerian polynomial*, and

$$f_{n+1}(x) = (xf_n(x))', \quad f_0(x) = \frac{1}{1-x}$$

Proposition 4.4. *The Eulerian numbers satisfy the recurrence*

$$A(n+1, k) = (k+1)A(n, k) + (n-k+1)A(n, k-1).$$

Proof. Clear denominators in $A_{n+1}(x) = (1-x)^{n+2} \frac{d}{dx} f_{n+1}(x)$ and extract $[x^k]$. $\square$

Corollary 4.5. *All Eulerian numbers are positive. They are displayed in the Eulerian triangle:*

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & & & 1 \\ & & 1 & & 4 & & 1 \\ & 1 & & 11 & & 11 & \\ 1 & & 26 & & 66 & & 26 & & 1 \end{array}$$

where each entry is the weighted sum of the two entries above it by the recurrence.

Corollary 4.6. $A(n, k) = A(n, n-1-k)$.

Proof. Induction with recurrence (4.4): the symmetry is preserved at each step. $\square$

Definition 4.7 (Descent). For $w \in S_n$, an index $i \in \{1, \dots, n-1\}$ is a *descent* of w if $w(i) > w(i+1)$. The *descent number* $\text{des}(w) = |\{i : w(i) > w(i+1)\}|$ is a statistic.

Theorem 4.8. $A(n, k)$ equals the number of permutations $w \in S_n$ with $\text{des}(w) = k$.

Proof. Let $A'(n, k)$ count permutations in S_n with k descents and check that it has the same recurrence by placing $n+1$ between elements (4.4). $\square$

Corollary 4.9. $A(n, k) = A(n, n-1-k)$, since the complement $w(i) \mapsto n+1-w(i)$ sends descents to ascents.

Corollary 4.10. $\sum_{k=0}^{n-1} A(n, k) = n!$, since every permutation in S_n has some number of descents.

Remark 4.11. Let $x_1, \dots, x_n \sim U(0, 1)$ be i.i.d. uniform random variables. Then

$$\mathbb{P}(\lfloor x_1 + \dots + x_n \rfloor = k) = \frac{A(n, k)}{n!}.$$

Lemma 4.12. *There is a bijection between S_n and increasing binary trees on vertex set $\{1, \dots, n\}$ (rooted binary trees where each vertex label is smaller than all labels in its subtree).*

Proof. Recursively: root is the minimum of the current segment of w ; entries left of it build the left subtree, entries right of it the right. In-order traversal inverts the map. $\square$

Theorem 4.13. $A(n, k)$ equals the number of increasing binary trees on $\{1, \dots, n\}$ with exactly k left edges (edges from a node to its left child).

Proof. Under the bijection to binary trees (4.12), left edges in the tree correspond to descents in the permutation (4.8). $\square$

Theorem 4.14. For a rooted plane binary tree T on n vertices, let f_T denote the number of increasing labelings of T (bijections $T \rightarrow \{1, \dots, n\}$ with each label smaller than its children). For each vertex v , let $h(v)$ denote the size of the subtree rooted at v . Then

$$f_T = \frac{n!}{\prod_v h(v)}.$$

Proof. Induct on n : label 1 is at the root; recurse on subtrees of sizes ℓ and $n-1-\ell$. Equivalently, a random bijection is proper with probability $1/\prod_v h(v)$. $\square$

4.2. Equidistribution.

Definition 4.15 (Equidistribution). Two statistics σ, μ on S_n are *equidistributed*, written $\sigma \sim \mu$, if

$$\forall k \in \mathbb{Z}_{\geq 0}, \quad |\{w \in S_n : \sigma(w) = k\}| = |\{w \in S_n : \mu(w) = k\}|$$

Definition 4.16 (Inversion). An *inversion* of $w \in S_n$ is a pair (i, j) with $i < j$ and $w(i) > w(j)$. The *inversion number* is $\text{inv}(w) = |\{(i, j) : i < j, w(i) > w(j)\}|$.

Definitions 4.17 (Eulerian and Mahonian statistics). A statistic is *Eulerian* if it is equidistributed with des , and *Mahonian* if it is equidistributed with inv .

Definition 4.18 (Number of cycles). $\text{cyc}(w)$ denotes the number of cycles in the cycle decomposition of $w \in S_n$, including fixed points.

Definition 4.19 (Major index). Let $\text{Des}(w) = \{i : w(i) > w(i+1)\}$ be the descent set. The *major index* of $w \in S_n$ is

$$\text{maj}(w) = \sum_{i \in \text{Des}(w)} i.$$

Theorem 4.20. $\text{maj} \sim \text{inv}$ on S_n .

Proof. See any of the following: [10], [4], [3] $\square$

Definition 4.21 (Left-to-right records). An index i is a *(left-to-right) record* of $w \in S_n$ if $w(i) > w(j)$ for all $j < i$ (i.e. $w(i)$ is a new maximum). The *record number* $\text{rec}(w) = |\{i : w(i) > \max_{j < i} w(j)\}|$, where $i = 1$ is always a record.

Definition 4.22 (Foata's first fundamental transformation). Given $w \in S_n$ in cycle notation (each cycle written with its largest element first, cycles sorted by increasing first element), the *Foata transformation* $\Phi(w)$ is obtained by erasing the parentheses and reading the result as one-line notation. Inverse is obtained by finding the left-to-right records.

Theorem 4.23. $\text{rec} \sim \text{cyc}$ on S_n .

Proof. Foata's Φ identifies left-to-right records with cycle leaders; bijectivity gives $\text{rec} \sim \text{cyc}$. $\square$

Definitions 4.24 (Excedance and anti-excedance). An index i is an *excedance* of $w \in S_n$ if $w(i) > i$, and an *anti-excedance* if $w(i) < i$. Set $\text{exc}(w) = |\{i : w(i) > i\}|$ and $\text{aexc}(w) = |\{i : w(i) < i\}|$.

Lemma 4.25. $\text{exc} \sim \text{aexc}$ on S_n .

Proof. $w \mapsto w^{-1}$ sends excedances of w to anti-excedances of w^{-1} . $\square$

Lemma 4.26. $\text{aexc} \sim \text{des}$ on S_n .

Proof. Foata's transformation Φ satisfies $\text{des}(\Phi(w)) = \text{aexc}(w)$. $\square$

Theorem 4.27. $\text{exc} \sim \text{des}$ on S_n .

Proof. $\text{exc} \sim \text{aexc}$ (4.25) and $\text{aexc} \sim \text{des}$ (4.26). $\square$

Definition 4.28 (Weak excedance). An index i is a *weak excedance* of $w \in S_n$ if $w(i) \geq i$. Set $\text{wexc}(w) = |\{i : w(i) \geq i\}|$.

Proposition 4.29. $\text{wexc}(w) \sim \text{exc}(w) + 1$ on S_n .

Proof. $\text{wexc}(w) = n - \text{aexc}(w) = n - \text{des}(\Phi(w))$ (4.26); for $\phi : w(i) \mapsto n + 1 - w(i)$, $\text{des}(\phi(w)) = n - 1 - \text{des}(w)$. Now 4.27. $\square$

5. POSETS

Definition 5.1 (Poset). A *partially ordered set* (poset) is a set P equipped with a binary relation $\leq$ satisfying: (i) $a \leq a$; (ii) $a \leq b$ and $b \leq a$ implies $a = b$; (iii) $a \leq b$ and $b \leq c$ implies $a \leq c$. Two elements are *incomparable* if neither $a \leq b$ nor $b \leq a$.

Notation 5.2. Write $a < b$ if $a \leq b$ and $a \neq b$, $a \geq b$ if $b \leq a$, and $a > b$ if $b < a$.

Definition 5.3 (Covering relation and Hasse diagram). The *covering relation* of P is $\{(a, b) : a < b\}$ where $a < b$ (read: b *covers* a) means that $a < b$ and there is no c with $a < c < b$. The *Hasse diagram* is the directed graph of the covering relation, drawn with b above a whenever $a < b$.

Definitions 5.4 (Chains and antichains). A *chain* is a totally ordered subset $a_1 < a_2 < \cdots < a_k$; it is a *k-chain* if it has k elements. A chain is *saturated* if $a_1 < a_2 < \cdots < a_k$. An *antichain* is a set of pairwise incomparable elements.

Definition 5.5 (Rank). A poset is *ranked* if there exists a function $\rho : P \rightarrow \mathbb{Z}_{\geq 0}$ such that $a < b$ implies $\rho(b) = \rho(a) + 1$. Let *rank number* $r_k = |\{a \in P : \rho(a) = k\}|$. Assume $r_0 > 0$.

Definitions 5.6 (Rank-symmetric, rank-unimodal, Sperner). A ranked poset of maximum rank n is:

- *rank-symmetric* if $r_k = r_{n-k}$ for all k ;
- *rank-unimodal* if $r_0 \leq r_1 \leq \cdots \leq r_k \geq \cdots \geq r_n$;
- *Sperner* if the maximum size of an antichain equals $\max_k r_k$.

5.1. Symmetric Chain Decomposition.

Definition 5.7 (Symmetric chain decomposition). A *symmetric chain decomposition* (SCD) of a ranked poset with maximum rank n is a partition of P into saturated chains $c_0 < c_1 < \dots < c_s$ such that $\rho(c_0) + \rho(c_s) = n$ for each chain.

Lemma 5.8. *If a ranked poset P has an SCD, then P is rank-symmetric, rank-unimodal, and Sperner.*

Proof. *Rank-symmetric:* opposite chain endpoints pair ranks k and $n - k$. *Rank-unimodal:* each chain hits every rank from $\rho(c_0)$ to $\rho(c_s)$ once, so r_k is a sum of interval indicators; symmetry of chains gives unimodality. *Sperner:* each chain has one rank- $\lfloor n/2 \rfloor$ vertex, so no antichain exceeds $r_{\lfloor n/2 \rfloor} = \max_k r_k$, and that rank is an antichain of that size. $\square$

Corollary 5.9. *The maximum size of an antichain in a poset P that has an SCD is $\max_k r_k = r_{\lfloor n/2 \rfloor}$.*

Proof. Since rank-symmetric and rank-unimodal, $\max_k r_k = r_{\lfloor n/2 \rfloor}$ (5.8). $\square$

Definition 5.10 (B_n). The *Boolean lattice* B_n is the poset of all subsets of $[n] = \{1, \dots, n\}$ ordered by inclusion. It is ranked by $\rho(S) = |S|$, with $r_k = \binom{n}{k}$ and maximum rank n .

Definition 5.11 (Product of posets). For posets P and Q , their *product* $P \times Q$ is the poset on the set $\{(p, q) : p \in P, q \in Q\}$ with $(p_1, q_1) \leq (p_2, q_2)$ if and only if $p_1 \leq_P p_2$ and $q_1 \leq_Q q_2$.

Lemma 5.12. *The product $[a] \times [b]$ of two chains has an SCD.*

Proof. Assume $a \leq b$. For $k = 1, \dots, a$ define $C_k = \{(k, j) : 1 \leq j \leq b - k + 1\} \cup \{(i, b - k + 1) : k < i \leq a\}$. Each step increases exactly one coordinate, so C_k is saturated. The endpoints are $(k, 1)$ and $(a, b - k + 1)$, whose ranks sum to $k + (a + b - k) = a + b$, hence C_k is symmetric. C_k partition $[a] \times [b]$. $\square$

Lemma 5.13. *If P and Q each have SCDs, then $P \times Q$ has an SCD.*

Proof. With SCDs $P = \bigsqcup_i C_i$ and $Q = \bigsqcup_j D_j$, we have $P \times Q = \bigsqcup_{i,j} C_i \times D_j$. Each factor is a product of two chains, hence has an SCD (5.12). Summing endpoints over C_i and D_j preserves $\rho(\text{start}) + \rho(\text{end}) = n_P + n_Q$. $\square$

Corollary 5.14. *Any product of chains has an SCD.*

Proof. Induct on the number of factors using 5.12 and 5.13. $\square$

Theorem 5.15 (De Bruijn's Theorem). *B_n has an SCD.*

Proof. $B_n \cong [2] \times [2] \times \dots \times [2]$ (n factors), identifying each subset $S \subseteq [n]$ with its indicator vector in $\{0, 1\}^n$. This product of chains has an SCD (5.14). $\square$

Theorem 5.16 (Sperner's Theorem). *The maximum size of an antichain in B_n is $\binom{n}{\lfloor n/2 \rfloor}$.*

Proof. B_n has an SCD (5.15); maximum antichain size is $r_{\lfloor n/2 \rfloor} = \binom{n}{\lfloor n/2 \rfloor}$ (5.9). $\square$

Corollary 5.17. *The maximum number of subsets of $[n]$ such that no one contains another is $\binom{n}{\lfloor n/2 \rfloor}$.*

Proof. Immediate from 5.16. Alternatively: find a bijective map from k -subsets to itself such that each subset doesn't intersect with its image. $\square$

5.2. Posets and Partitions.

Theorem 5.18 (Greene's Theorem for Posets). *For any poset P , let $\ell_k(P)$ be the maximum size of a union of k chains, and $m_k(P)$ the maximum size of a union of k antichains. Define partitions*

$$\lambda(P) = (\ell_1, \ell_2 - \ell_1, \ell_3 - \ell_2, \dots), \quad \mu(P) = (m_1, m_2 - m_1, m_3 - m_2, \dots).$$

Then $\lambda(P)$ and $\mu(P)$ are conjugate partitions: $\mu(P) = \lambda(P)'$.

Proof. See [6] □

Definition 5.19 (Poset of a permutation). For $w \in S_n$, define the poset P_w on $\{x_1, \dots, x_n\}$ by $x_i \leq x_j$ if $i \leq j$ and $w(i) \leq w(j)$.

Lemma 5.20. *Chains in P_w correspond exactly to increasing subsequences of w .*

Proof. A chain $x_{i_1} < x_{i_2} < \dots < x_{i_k}$ in P_w requires $i_1 < i_2 < \dots < i_k$ and $w(i_1) < w(i_2) < \dots < w(i_k)$, which is exactly an increasing subsequence of w of length k . □

Lemma 5.21. $\lambda(P_w) = (\ell_1, \ell_2 - \ell_1, \dots)$, equals the Schensted shape of w .

Proof. $\ell_k(P_w)$ is the maximum size of a union of k increasing subsequences of w (5.20). $\ell_1 + \dots + \ell_k = \lambda_1 + \dots + \lambda_k$ where $\lambda = \lambda(w)$ is the Schensted shape (3.23). □

Theorem 5.22 (Dilworth's Theorem). *The maximum size of an antichain in P equals the minimum number of chains needed to cover P .*

Proof. By 5.18, $m_1(P) = \lambda'_1(P)$ equals the minimum chain-cover number. Also, a k -chain cover meets any antichain in at most k points. □

Theorem 5.23 (Mirsky's Theorem). *The maximum length of a chain in P equals the minimum number of antichains needed to cover P .*

Proof. By 5.18, $\ell_1(P) = \lambda_1(P) = \mu'_1(P)$ is the minimum antichain-cover number; a k -antichain cover meets any chain in at most k points. □

Definition 5.24 (Order ideals). An *order ideal* of P is a subset $I \subseteq P$ such that $a \in I$ and $b \leq a$ implies $b \in I$. The set $J(P)$ of all order ideals of P , ordered by inclusion, is the *lattice of order ideals* of P .

Definition 5.25 (Young's lattice). The *Young lattice* $L_{m,n} = J([m] \times [n])$ consists of partitions $\lambda \subseteq m \times n$ ordered by diagram inclusion, ranked by $|\lambda|$. When m and n are infinite, denote the corresponding Young lattice by $\mathbb{Y}$.

Theorem 5.26. *The lattice $J([m] \times [n])$ has a symmetric chain decomposition.*

Proof. See [7]. □

Corollary 5.27. *Young's lattice $L_{m,n}$ is rank-symmetric and rank-unimodal.*

Proof. $J([m] \times [n])$ has an SCD; it is rank-symmetric and rank-unimodal (5.8). □

6. q -ANALOGS

Definition 6.1 (q -analogs). A q -analog of a numerical quantity is a polynomial or rational function in q that specializes to the original quantity at $q = 1$.

Definition 6.2 (q -number). For $n \geq 0$, the q -number is

$$[n]_q = 1 + q + \cdots + q^{n-1} = \frac{1 - q^n}{1 - q}.$$

Definition 6.3 (q -factorial). For $n \geq 0$, the q -factorial is

$$[n]!_q = [1]_q [2]_q \cdots [n]_q.$$

Theorem 6.4.

$$[n]!_q = \sum_{w \in S_n} q^{\text{inv}(w)} = \sum_{w \in S_n} q^{\text{maj}(w)}.$$

Proof. Induct on n . Build permutations in S_n by inserting n into a permutation of S_{n-1} . If n is inserted in position i , the inversion number increases by $n - i$; summing over all i gives a factor of $[n]_q$. Use 4.20 for maj. $\square$

Definition 6.5 (q -binomial coefficient). For $0 \leq k \leq n$, the q -binomial coefficient is

$$\binom{n}{k}_q = \frac{[n]!_q}{[k]!_q [n-k]!_q}.$$

Corollary 6.6.

$$\binom{n}{k}_q = \binom{n}{n-k}_q.$$

Proof. Immediate from the definition. $\square$

Proposition 6.7 (q -Pascal formula). For $1 \leq k \leq n$,

$$\binom{n}{k}_q = \binom{n-1}{k}_q + q^{n-k} \binom{n-1}{k-1}_q.$$

Equivalently,

$$\binom{n}{k}_q = q^k \binom{n-1}{k}_q + \binom{n-1}{k-1}_q.$$

Proof. Use the definition and $[n]_q = [n-k]_q + q^{n-k}[k]_q$. $\square$

Corollary 6.8. For all $0 \leq k \leq n$, $\binom{n}{k}_q$ is a polynomial in q with positive integer coefficients. These coefficients are arranged by the q -Pascal triangle with 6.7.

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & & & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 1+q & & 1 \\
 & & & & & & \\
 & 1 & & 1+q+q^2 & & 1+q+q^2 & 1 \\
 & & & & & & \\
 1 & 1+q+q^2+q^3 & & 1+q+2q^2+q^3+q^4 & & 1+q+q^2+q^3 & 1
 \end{array}$$

Proof. Induct on n using 6.7. $\square$

6.1. Grassmannian.

Definition 6.9 (Grassmannian over $\mathbb{F}_q$). Let q be a prime power. The *Grassmannian* $\text{Gr}(k, n, \mathbb{F}_q)$ is the set of k -dimensional linear subspaces of $\mathbb{F}_q^n$.

Theorem 6.10. $|\text{Gr}(k, n, \mathbb{F}_q)| = \binom{n}{k}_q$.

Proof. Count full-rank $k \times n$ matrices over $\mathbb{F}_q$. Such a matrix determines a k -dimensional row space, and two matrices determine the same subspace exactly when they differ by left multiplication by an element of $GL_k(\mathbb{F}_q)$. Divide the number of full-rank $k \times n$ matrices by $|GL_k(\mathbb{F}_q)|$; count: $\frac{(q^n - 1) \cdots (q^n - q^{k-1})}{(q^k - 1) \cdots (q^k - q^{k-1})}$ $\square$

Theorem 6.11.

$$|\text{Gr}(k, n, \mathbb{F}_q)| = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}.$$

Proof. RREF is unique: pivots give a path on the boundary of some $\lambda \subseteq k \times (n-k)$, and free entries fill the cells of λ , so shape λ contributes $q^{|\lambda|}$ subspaces. $\square$

6.2. q -Binomial Coefficient and Partitions.

Theorem 6.12.

$$\binom{n}{k}_q = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|},$$

where the sum is over all Young diagrams λ contained in the $k \times (n-k)$ rectangle.

Proof. Split by whether λ uses the bottom row to match q -Pascal and initial data. Or combine 6.10 and 6.11. $\square$

Corollary 6.13.

$$\sum_{\substack{\lambda \\ \lambda_1 \leq n-k \\ \lambda'_1 \leq k}} q^{|\lambda|} = \binom{n}{k}_q.$$

Letting $n \rightarrow \infty$,

$$\sum_{\lambda'_1 \leq k} q^{|\lambda|} = \sum_{\lambda_1 \leq k} q^{|\lambda|} = \frac{1}{(1-q)(1-q^2) \cdots (1-q^k)}.$$

Proof. The first statement is 6.12. Passing to the limit removes the width restriction and leaves partitions with at most k rows. Column version is achieved by conjugating partitions. $\square$

Corollary 6.14.

$$\sum_{\lambda} q^{|\lambda|} = \prod_{i=1}^{\infty} \frac{1}{1-q^i}.$$

Proof. Let $k \rightarrow \infty$ in 6.13. $\square$

Corollary 6.15. The coefficients of $\binom{n}{k}_q$ are symmetric and unimodal.

Proof. $\binom{n}{k}_q$ is the rank generating function of Young's lattice $L_{k, n-k}$ (6.12). $L_{k, n-k}$ is rank-symmetric and rank-unimodal (5.27). $\square$

7. PARTITION FUNCTION

Definition 7.1 (Partition function). The *partition function* $p(n)$ is the number of partitions of n .

Remark 7.2. As $n \rightarrow \infty$,

$$p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}.$$

This is the Hardy–Ramanujan asymptotic formula.

7.1. Generating Functions for Partitions.

Notation 7.3 (Multiplicity notation for partitions). Write a partition λ as

$$\lambda = (m_1, m_2, m_3 \dots),$$

where m_i is the number of parts of λ equal to i . Then

$$|\lambda| = \sum_{i \geq 1} i m_i.$$

Lemma 7.4. For partitions with at most k parts,

$$\sum_{\lambda'_1 \leq k} x^{|\lambda|} = \frac{1}{(1-x)(1-x^2) \cdots (1-x^k)}.$$

Equivalently, for partitions with at most k columns,

$$\sum_{\lambda_1 \leq k} x^{|\lambda|} = \frac{1}{(1-x)(1-x^2) \cdots (1-x^k)}.$$

Hence for all partitions,

$$\sum_{\lambda} x^{|\lambda|} = \prod_{i=1}^{\infty} \frac{1}{1-x^i}.$$

Proof. Multiplicities $m_i \geq 0$ contribute x^{im_i} , so

$$\sum_{\lambda} x^{|\lambda|} = \prod_{i=1}^{\infty} (1 + x^i + x^{2i} + \cdots) = \prod_{i=1}^{\infty} \frac{1}{1-x^i}.$$

Truncate at $i \leq k$ for the bounded case; conjugate for columns. Also 6.13, 6.14. $\square$

Proposition 7.5. The generating function of the partition function is

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{i=1}^{\infty} \frac{1}{1-x^i}.$$

Proof. By 7.4, the coefficient of x^n in $\prod_{i=1}^{\infty} (1-x^i)^{-1}$ is exactly the number of partitions of n . $\square$

Definition 7.6 (Partitions with distinct parts). Let $p_{\text{dist}}(n)$ denote the number of partitions of n into distinct parts.

Definition 7.7 (Partitions with odd parts). Let $p_{\text{odd}}(n)$ denote the number of partitions of n into odd parts.

Theorem 7.8 (Euler's Partition Theorem). For all $n \geq 0$,

$$p_{\text{dist}}(n) = p_{\text{odd}}(n).$$

Proof. The generating function for partitions into distinct parts is

$$\sum_{n=0}^{\infty} p_{\text{dist}}(n)x^n = \prod_{i=1}^{\infty} (1 + x^i),$$

while the generating function for partitions into odd parts is

$$\sum_{n=0}^{\infty} p_{\text{odd}}(n)x^n = \prod_{i=1}^{\infty} \frac{1}{1 - x^{2i-1}}.$$

Since

$$\prod_{i=1}^{\infty} (1 + x^i) = \prod_{i=1}^{\infty} \frac{1 - x^{2i}}{1 - x^i} = \prod_{i=1}^{\infty} \frac{1}{1 - x^{2i-1}},$$

the generating functions are equal. $\square$

7.2. Recurrence Relation for Partitions.

Definition 7.9 (Pentagonal numbers). For $k \in \mathbb{Z}$, the *generalized pentagonal numbers* are

$$\frac{k(3k-1)}{2}.$$

Equivalently, they are the numbers

$$0, 1, 2, 5, 7, 12, 15, \dots$$

obtained from $k = 0, \pm 1, \pm 2, \pm 3, \dots$

Lemma 7.10 (Franklin's involution). *On partitions of n into distinct parts there is an involution ϕ that flips the parity of the number of parts. Its fixed points are*

$$(2k-1, 2k-2, \dots, k) \quad n = \frac{k(3k-1)}{2}, \quad (2k, 2k-1, \dots, k+1) \quad n = \frac{k(3k+1)}{2},$$

each having k parts; ϕ has no fixed points for other n .

Proof. For $\lambda = (\lambda_1 > \dots > \lambda_k)$, let $s = \lambda_k$ and let t be the largest j with $\lambda_i = \lambda_1 - (i-1)$ for $i \leq j$. If $s \leq t$ and $s < k$, define $\phi(\lambda)$ by adding 1 to each of the top s parts and deleting the smallest part; if $s > t$, by subtracting 1 from each of the top t parts and appending a new smallest part of size t . The two operations are inverse and change k by ± 1 . Both fail exactly when the top staircase meets the bottom, i.e. $s = t = k$ or $s = t + 1 = k + 1$, giving the listed fixed points. $\square$

Lemma 7.11. *Let $a(n)$ be the number of partitions of n into distinct parts with an even number of parts, and let $b(n)$ be the number of partitions of n into distinct parts with an odd number of parts. Then*

$$a(n) - b(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{k(3k \pm 1)}{2} \text{ for some } k \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. By 7.10, partitions of opposite parity cancel except at fixed points; for $n = k(3k \pm 1)/2$ the unique fixed point has k parts and contributes $(-1)^k$. $\square$

Lemma 7.12.

$$\prod_{i=1}^{\infty} (1 - x^i) = \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2}.$$

Equivalently,

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k-1)/2} + x^{k(3k+1)/2} \right).$$

Proof. Expanding the left-hand side, the coefficient of x^n is exactly $a(n) - b(n)$, where $a(n), b(n)$ are as in 7.11. Apply 7.11. Equivalently, specialize 7.15 with $x = q^{3/2}$ and $y^2 = -q^{1/2}$. $\square$

Theorem 7.13 (Euler's Pentagonal Number Theorem). *Let $p(0) = 1$ and $p(n) = 0$ for $n < 0$. Then for $n \geq 1$,*

$$p(n) = \sum_{k=1}^{\infty} (-1)^{k-1} \left(p\left(n - \frac{k(3k-1)}{2}\right) + p\left(n - \frac{k(3k+1)}{2}\right) \right).$$

Proof. From 7.5 and 7.12,

$$1 = \left(\sum_{n=0}^{\infty} p(n)x^n \right) \left(\sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2} \right).$$

Compare coefficients of x^n . $\square$

7.3. Jacobi Triple Product.

Lemma 7.14. *Fix $a, b \in \mathbb{Z}_{\geq 0}$, set $k = a - b$, and fix $\ell \in \mathbb{Z}_{\geq 0}$. There is a bijection between pairs (μ, ν) and partitions λ such that μ is a partition into a distinct positive parts, ν is a partition into b distinct positive parts,*

$$|\mu| + |\nu| - b = \ell,$$

and

$$|\lambda| = \ell - \frac{k(k+1)}{2}.$$

Proof. Write $\delta_m = (m-1, m-2, \dots, 1, 0)$ for the staircase of length m , so $|\delta_m| = \binom{m}{2}$, and write λ' for conjugation of partitions.

By subtraction, create $\tilde{\mu} = \mu - \delta_a$ and $\tilde{\nu} = \nu - \delta_b$. Glue the start of the last column of $\tilde{\nu}'$'s Young diagram with the start of the last row of $\tilde{\mu}$'s Young diagram and discard the remaining staircase parts to form a single partition λ .

Given λ and $k = a - b \in \mathbb{Z}$, the location of the cut between the $\tilde{\mu}$ -block and the $\tilde{\nu}'$ -block is recovered by attaching the staircase

$$\begin{cases} \delta_{k+1} \text{ to the left of } \lambda, & k \geq 0, \\ \delta_{-k} \text{ to the top of } \lambda, & k < 0. \end{cases}$$

This pins the unique horizontal edge along which λ splits; transposing the upper block and re-adding δ_a, δ_b recovers (μ, ν) . $\square$

Theorem 7.15 (Jacobi Triple Product).

$$\prod_{n=1}^{\infty} (1 - x^{2n})(1 + x^{2n-1}y^2)(1 + x^{2n-1}y^{-2}) = \sum_{k=-\infty}^{\infty} x^{k^2} y^{2k}.$$

Proof. Write $x^2 = q$ and $y^2 = zq^{1/2}$; equivalently prove

$$\prod_{n=1}^{\infty} (1 - q^n)(1 + zq^n)(1 + q^{n-1}z^{-1}) = \sum_{k=-\infty}^{\infty} z^k q^{k(k+1)/2},$$

which rearranges to

$$\left(\prod_{n=1}^{\infty} (1 + zq^n) \right) \left(\prod_{n=1}^{\infty} (1 + q^{n-1}z^{-1}) \right) = \left(\prod_{n=1}^{\infty} \frac{1}{1 - q^n} \right) \left(\sum_{k=-\infty}^{\infty} z^k q^{k(k+1)/2} \right).$$

The coefficient of z^a in $\prod_{n \geq 1} (1 + zq^n)$ is $\sum_{\mu} q^{|\mu|}$ over partitions μ into a distinct parts; the coefficient of z^{-b} in $\prod_{n \geq 1} (1 + q^{n-1}z^{-1})$ is $\sum_{\nu} q^{|\nu| - b}$ over partitions ν into b distinct parts. Thus the $q^{\ell} z^k$ coefficient on the left counts pairs (μ, ν) of partitions into distinct parts with $|\mu| + |\nu| - b = \ell$ and $a - b = k$. On the right, $\prod_{n \geq 1} (1 - q^n)^{-1}$ generates all partitions, so the $q^{\ell} z^k$ coefficient counts pairs (λ, k) with $|\lambda| = \ell - \frac{k(k+1)}{2}$. Lemma 7.14 identifies these counts. $\square$

Corollary 7.16.

$$\prod_{n=1}^{\infty} (1 - x^n)^3 = \sum_{k=0}^{\infty} (-1)^k (2k+1) x^{k(k+1)/2}.$$

Proof. Specialize 7.15 with $x = -q^{1/2}$ and $y = z^{1/2}q^{1/4}$; derive the identity with $(1 - z)$ and evaluate at $z = 1$. Grouping terms by $k(k+1)/2$ yields the coefficient of $x^{k(k+1)/2}$. $\square$

Corollary 7.17.

$$\prod_{n=1}^{\infty} \frac{1 - x^n}{1 + x^n} = \sum_{k=-\infty}^{\infty} (-1)^k x^{k^2}.$$

Proof. Specialize 7.15 with $x = -q$ and $y = 1$. $\square$

Proposition 7.18.

$$\prod_{n=1}^{\infty} (1 + xq^n) = \sum_{k=0}^{\infty} \frac{q^{k(k+1)/2} x^k}{(1 - q)(1 - q^2) \cdots (1 - q^k)}.$$

Proof. Expand the left-hand side using partitions and subtract δ_k from the partitions; use 7.4 to divide partitions by their lengths. $\square$

8. DIFFERENTIAL POSETS

Proposition 8.1. *Rank number of Young's lattice at n is $p(n)$.*

Proof. Since the rank is given by the size of the partition, the rank number of Young's lattice is the number of partitions of n . $\square$

Proposition 8.2. *For $\lambda \vdash n$, $f^{\lambda} = |\text{SYT}(\lambda)|$ equals the number of saturated chains $\emptyset = \lambda^{(0)} < \cdots < \lambda^{(n)} = \lambda$ in $\mathbb{Y}$. Hence*

$$\sum_{|\lambda|=n} (f^{\lambda})^2 = n!$$

counts paths in the Hasse diagram of $\mathbb{Y}$ that ascend n covers from $\emptyset$ and then descend n covers back to $\emptyset$.

Proof. The map $T \mapsto (\lambda^{(0)}, \dots, \lambda^{(n)})$ with $\lambda^{(i)} = \{c \in T : T(c) \leq i\}$ is a bijection between $\text{SYT}(\lambda)$ and saturated $\emptyset$ -to- λ chains in $\mathbb{Y}$, since each cover $\lambda^{(i-1)} \lessdot \lambda^{(i)}$ adds exactly one cell labeled with i . Thus

$$\sum_{|\lambda|=n} (f^\lambda)^2 = \sum_{|\lambda|=n} |\text{SYT}(\lambda)|^2 = |\{(P, Q) : P, Q \in \text{SYT}(\lambda), \lambda \vdash n\}| = |S_n| = n!,$$

where the third equality is 3.13. Concatenating the chain for P with the reverse of the chain for Q yields the asserted ascend-then-descend path. $\square$

Definition 8.3. Let V be the $\mathbb{R}$ -vector space with basis $\{v_\lambda : \lambda \in \mathbb{Y}\}$, graded by $V_n = \text{span}\{v_\lambda : |\lambda| = n\}$, so $V = \bigoplus_{n \geq 0} V_n$. Define linear $U, D : V \rightarrow V$ by

$$U(v_\lambda) = \sum_{\mu \succ \lambda} v_\mu, \quad D(v_\lambda) = \sum_{\mu \lessdot \lambda} v_\mu, \quad D(v_\emptyset) = 0,$$

where $\succ$ and $\lessdot$ denote cover relations in $\mathbb{Y}$.

Lemma 8.4. On V , $DU - UD = I$.

Proof. Equivalent combinatorially to: (i) for every λ , $|\{\mu : \mu \succ \lambda\}| = 1 + |\{\mu : \mu \lessdot \lambda\}|$; (ii) for distinct λ, μ of the same size, the number of ν with $\nu \lessdot \lambda, \mu$ equals the number of ν with $\nu \succ \lambda, \mu$. $\square$

Theorem 8.5. For all $n \geq 0$, $D^n U^n(v_\emptyset) = n! v_\emptyset$.

Proof. First prove $DU^n(v_\emptyset) = n U^{n-1}(v_\emptyset)$ by induction on n , using $DU = UD + I$ from 8.4. Then induct on n for $D^n U^n(v_\emptyset)$, writing $D^n U^n = D^{n-1}(DU^n)$. $\square$

Remark 8.6. On smooth functions, $\frac{d}{dx}$ and multiplication by x satisfy $\frac{d}{dx} x - x \frac{d}{dx} = I$ and $\frac{d^n}{dx^n} x^n = n!$.

Definition 8.7 (Differentiable poset). A locally finite graded poset P with minimum $\hat{0}$ is *differentiable* if its up and down operators (defined as for $\mathbb{Y}$) satisfy $DU - UD = I$, equivalently: (i) for every $\lambda \in P$, $|\{\mu : \mu \succ \lambda\}| = 1 + |\{\mu : \mu \lessdot \lambda\}|$; (ii) for distinct λ, μ of equal rank, $|\{\nu : \nu \lessdot \lambda, \mu\}| = |\{\nu : \nu \succ \lambda, \mu\}|$.

Theorem 8.8. Let P be differentiable and let $c_n(\lambda)$ be the number of saturated chains from $\hat{0}$ to λ when $\text{rank } \rho(\lambda) = n$. Then

$$\sum_{\rho(\lambda)=n} c_n(\lambda)^2 = n!.$$

Proof. Repeat the proof of 8.5 in the operator algebra of P . $\square$

Example 8.9 (Fibonacci lattice). A differentiable poset F has one element at rank 0 and one at rank 1. For $k \geq 2$, elements of rank k arise from ranks $k-1$ and $k-2$: each λ of rank $k-1$ has a unique cover λ' of rank k (the only relation $\cdot \lessdot \lambda'$ into rank k), and each μ of rank $k-2$ has a unique cover μ' of rank k such that for all λ of rank $k-1$, one has $\lambda \lessdot \mu'$ if and only if $\mu \lessdot \lambda$.

8.1. Oscillating Tableaux.

Definition 8.10. A U/D -word of length $2n$ is $W = W_1 \cdots W_{2n} \in \{U, D\}^{2n}$ with n U 's and n D 's such that every prefix has at least as many U 's as D 's.

Definition 8.11. Fix a differentiable poset P with minimum $\hat{0}$. For a U/D -word W , set

$$f_W = \left| \left\{ (\lambda^{(0)}, \dots, \lambda^{(2n)}) \in P^{2n+1} : \lambda^{(0)} = \lambda^{(2n)} = \hat{0}, \right. \right. \\ \left. \left. \lambda^{(i-1)} \leq \lambda^{(i)} \text{ if } W_i = U, \lambda^{(i)} \leq \lambda^{(i-1)} \text{ if } W_i = D \right\} \right|.$$

A path of this kind is an *oscillating tableau* with profile W .

Definition 8.12. For a U/D -word W of length $2n$, let $K_W = (K_1, \dots, K_n) \subseteq n \times n$ be the partition whose southeast boundary inside the $n \times n$ box is traced by W , with U a horizontal step and D a vertical step.

Lemma 8.13. $W \mapsto K_W$ bijects U/D -words of length $2n$ with partitions $K \subseteq n \times n$ satisfying $K_{n-i} > i$ for $i = 0, \dots, n-1$; equivalently, K contains the staircase $(n, n-1, \dots, 1)$.

Proof. The boundary path inside the $n \times n$ box has n horizontal and n vertical steps. The Dyck condition translates to $K_{n-i} > i$ at each row $n-i$. $\square$

Definition 8.14. r_W is the number of placements of n non-attacking rooks in K_W (no two rooks share a row or a column).

Theorem 8.15. For any partition $K = (K_1, \dots, K_n) \subseteq n \times n$ with $K_{n-i} > i$,

$$r_W = \prod_{i=0}^{n-1} (K_{n-i} - i) = \prod_{i=0}^{n-1} (K'_{n-i} - i),$$

where K' is the conjugate partition.

Proof. Place rooks row by row from row n upward: row $n-i$ has K_{n-i} cells, of which i columns are already occupied, leaving $K_{n-i} - i \geq 1$ choices. The conjugate form follows by applying the same argument to columns. $\square$

Theorem 8.16. Let P be differentiable. If $W = W_1 U D W_2$, $W' = W_1 D U W_2$, and $W'' = W_1 W_2$, then

$$f_W = f_{W'} + f_{W''}.$$

Proof. Use $DU - UD = I$. $\square$

Theorem 8.17. $f_W = r_W$.

Proof. Induct on n . Choose any UD -substring of W ; 8.16 gives $f_W = f_{W'} + f_{W''}$. The same recurrence $r_W = r_{W'} + r_{W''}$ holds for rook counts: in K_W , the UD step traces a unique outer corner c , and rook placements split by occupancy of c . Placements with a rook on c correspond to rook placements of $n-1$ rooks on $K_{W''} \subseteq (n-1) \times (n-1)$; those without correspond to rook placements on $K_{W'} = K_W \setminus \{c\}$. Base case $W = \emptyset$: $f_\emptyset = r_\emptyset = 1$. $\square$

9. CAYLEY'S FORMULA

Definition 9.1 (Labeled tree). A *labeled tree on $[n]$* is a tree with vertex set $[n] = \{1, 2, \dots, n\}$.

Lemma 9.2. *If $g \in \mathbb{R}[x_1, \dots, x_n]$ has $\deg g < n$ and $g|_{x_i=0} \equiv 0$ for every $i \in [n]$, then $g \equiv 0$.*

Proof. Any monomial $\prod_i x_i^{a_i}$ in g has $\sum_i a_i < n$, hence some $a_j = 0$, contradicting vanishing on $x_j = 0$. $\square$

Theorem 9.3 (Multivariate Generalized Cayley's Formula). *For*

$$F_n(x_1, \dots, x_n) = \sum_T \prod_{i=1}^n x_i^{\deg_T(i)-1},$$

summing over labeled trees T on $[n]$,

$$F_n(x_1, \dots, x_n) = (x_1 + \dots + x_n)^{n-2}.$$

Proof. By handshaking, $\sum_i (\deg_T(i) - 1) = 2(n-1) - n = n-2$, so F_n is homogeneous of degree $n-2$. Set $G_n := F_n - (x_1 + \dots + x_n)^{n-2}$ and induct on n , with $G_2 \equiv 0$ as base. By symmetry it suffices to verify $G_n(x_1, \dots, x_{n-1}, 0) \equiv 0$. Only trees with $\deg_T(n) = 1$ contribute: n is a leaf adjacent to some $j \in [n-1]$, and deleting n leaves a tree T' on $[n-1]$. Hence

$$G_n(x_1, \dots, x_{n-1}, 0) = \left(\sum_{j=1}^{n-1} x_j \right) F_{n-1}(x_1, \dots, x_{n-1}) - (x_1 + \dots + x_{n-1})^{n-2}.$$

By induction $F_{n-1} = (x_1 + \dots + x_{n-1})^{n-3}$, so the right side vanishes. Apply 9.2. $\square$

Corollary 9.4. *For $d_1, \dots, d_n \geq 1$ with $\sum_i d_i = 2(n-1)$, the number of labeled trees on $[n]$ with $\deg_T(i) = d_i$ is $\binom{n-2}{d_1-1, \dots, d_n-1} = \frac{(n-2)!}{\prod_{i=1}^n (d_i-1)!}$.*

Proof. Extract the coefficient of $x_1^{d_1-1} \dots x_n^{d_n-1}$ in 9.3. $\square$

Theorem 9.5 (Cayley's Formula). *For $n \geq 2$, the number of labeled trees on $[n]$ is n^{n-2} .*

Proof. Specializing 9.3 with $x_i = 1$ for all i gives $F_n(1, \dots, 1) = (n-1)^{n-2}$. $\square$

9.1. Prüfer's Coding.

Construction 9.6 (Prüfer code). For a labeled tree T on $[n]$, repeat $n-2$ times: let ℓ be the minimal leaf, record its unique neighbor c , and delete ℓ . The output $\text{code}(T) = (c_1, \dots, c_{n-2}) \in [n]^{n-2}$. Vertex i appears $\deg_T(i) - 1$ times in $\text{code}(T)$; equivalently, i is a leaf of T iff i does not appear in $\text{code}(T)$.

Theorem 9.7. *The map $T \mapsto \text{code}(T)$ is a bijection between labeled trees on $[n]$ and $[n]^{n-2}$.*

Proof. Construct the inverse explicitly. Given $(c_1, \dots, c_{n-2}) \in [n]^{n-2}$, set $C \leftarrow (c_1, \dots, c_{n-2})$ and $L \leftarrow [n]$, then repeat $n-2$ times: take the minimal $\ell \in L$ not appearing in C , add the edge $\{\ell, c\}$ where c is the first entry of C , and remove ℓ from L and the first entry from C ; finally, join the two remaining elements of L by an edge. The output is a tree on $[n]$, and the two procedures invert each other step by step. $\square$

Corollary 9.8. For $d_1, \dots, d_n \geq 1$ with $\sum_i d_i = 2(n-1)$, $T \mapsto \text{code}(T)$ is a bijection between labeled trees on $[n]$ with $\deg_T(i) = d_i$ and sequences in $[n]^{n-2}$ in which i appears exactly $d_i - 1$ times. Explicitly, $(d_1-1, \dots, d_n-1) = \frac{(n-2)!}{\prod_{i=1}^n (d_i-1)!}$.

Remark 9.9. Counting all $[n]^{n-2}$ sequences gives n^{n-2} trees, recovering 9.5.

9.2. Egecioğlu–Remmel’s Bijection.

Theorem 9.10 (Egecioğlu–Remmel’s Bijection). Labeled trees on $[n]$ are in bijection with functions $f : [n] \rightarrow [n]$ satisfying $f(1) = 1$ and $f(n) = n$.

Proof. Given T , let $1 = a_1, \dots, a_k = n$ be the path from 1 to n in T . The right-to-left minima include $a_1 = 1$ and $a_k = n$. Cut the path after each right-to-left minimum into blocks; convert each block into a directed cycle, and direct every off-path edge toward its parent on the path. The resulting functional digraph defines $f : [n] \rightarrow [n]$ with $f(1) = 1$ and $f(n) = n$. $\square$

Remark 9.11. $|\{f : [n] \rightarrow [n] : f(1) = 1, f(n) = n\}| = n^{n-2}$, recovering 9.5.

10. KIRCHHOFF’S MATRIX–TREE THEOREM

Definition 10.1 (Spanning tree). For a finite multigraph $G = (V, E)$ without loops, a *spanning tree* is a subgraph (V, E') with $E' \subseteq E$ that is a tree on V . Write $\tau(G)$ for the number of spanning trees of G .

Definition 10.2 (Adjacency matrix). For G on $V = [n]$, the *adjacency matrix* $A = (a_{ij}) \in \mathbb{Z}_{\geq 0}^{n \times n}$ is given by $a_{ij} = |\{e \in E : i \in e, j \in e\}|$, with $a_{ii} = 0$ since G has no loops.

Definition 10.3 (Incidence matrix). For G with $|V| = n$ and $|E| = m$, the (unsigned) *incidence matrix* is $C = (c_{ij}) \in \{0, 1\}^{n \times m}$ with $c_{ij} = 1$ iff vertex i is incident to edge j .

Definition 10.4 (Laplacian, reduced Laplacian). With $D = \text{diag}(d_1, \dots, d_n)$, $d_i = \deg_G(i)$, the *Laplacian* of G is $L = D - A$. For fixed $i_0 \in [n]$, the *reduced Laplacian* $\tilde{L} \in \mathbb{Z}^{(n-1) \times (n-1)}$ is obtained by deleting row i_0 and column i_0 of L .

Remark 10.5. Every row sum and column sum of L equals zero, so $\det L = 0$.

Definition 10.6 (Oriented incidence matrix). After fixing an arbitrary orientation of each edge, the *oriented incidence matrix* is $B = (b_{ij}) \in \{-1, 0, 1\}^{n \times m}$ with $b_{ij} = 1$ if i is the source of edge j , $b_{ij} = -1$ if i is the target of edge j , and $b_{ij} = 0$ otherwise. Let $\tilde{B} \in \{-1, 0, 1\}^{(n-1) \times m}$ denote the matrix obtained by deleting row i_0 of B .

Lemma 10.7. $L = BB^T$ and $\tilde{L} = \tilde{B}\tilde{B}^T$.

Proof. $(BB^T)_{ii}$ counts the nonzero entries in row i of B , equal to d_i . For $i \neq j$, each edge between i and j contributes $1 \cdot (-1) = -1$ to $(BB^T)_{ij}$ regardless of orientation, so $(BB^T)_{ij} = -a_{ij}$. Hence $BB^T = D - A = L$. Restricting both factors to rows $\neq i_0$ yields $\tilde{B}\tilde{B}^T = \tilde{L}$. $\square$

Theorem 10.8 (Cauchy–Binet Formula). For $M \in \mathbb{R}^{k \times m}$, $N \in \mathbb{R}^{m \times k}$ with $k \leq m$,

$$\det(MN) = \sum_{\substack{S \subseteq [m] \\ |S|=k}} \det(M_S) \det(N_S),$$

where M_S is the $k \times k$ submatrix on column set S and N_S the $k \times k$ submatrix on row set S .

Proof. Proofs can be found in papers including [9]. $\square$

Lemma 10.9. For $S \subseteq E$ with $|S| = n - 1$,

$$\det \tilde{B}_S = \begin{cases} \pm 1 & S \text{ is the edge set of a spanning tree of } G, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. If S is not a spanning tree, the subgraph (V, S) contains an undirected cycle C . Signing the columns of B indexed by $E(C)$ according to a fixed traversal produces a linear dependence relation, hence the same dependence in $\tilde{B}_S$, so $\det \tilde{B}_S = 0$.

If $S = E(T)$ for a spanning tree T , since every tree on $n \geq 2$ vertices has at least two leaves, T has a leaf $i \neq i_0$. Row i of $\tilde{B}_S$ has exactly one nonzero entry, ± 1 . Cofactor expansion along that row gives $\det \tilde{B}_S = \pm \det \tilde{B}_{S'}$ for $S' = E(T - i)$, a spanning tree of G/i on $n - 1$ vertices. Induction on n , with base case $n = 2$ trivial, gives $\det \tilde{B}_S = \pm 1$. $\square$

Theorem 10.10 (Kirchhoff's Matrix-Tree Theorem). $\tau(G) = \det \tilde{L}$, independent of the choice of i_0 .

Proof. By 10.7 and 10.8,

$$\det \tilde{L} = \det(\tilde{B}\tilde{B}^\top) = \sum_{\substack{S \subseteq E \\ |S|=n-1}} (\det \tilde{B}_S)^2,$$

and 10.9 reduces the right-hand side to $\tau(G)$. $\square$

Remark 10.11. For $G = K_n$, $L = nI_n - \mathbf{1}\mathbf{1}^\top$ and $\tilde{L} = nI_{n-1} - \mathbf{1}\mathbf{1}^\top$. The matrix $\mathbf{1}\mathbf{1}^\top \in \mathbb{R}^{(n-1) \times (n-1)}$ has rank 1 with trace $n - 1$, so its eigenvalues are 0 ($n - 2$ times) and $n - 1$. Hence $\tilde{L}$ has eigenvalues n ($n - 2$ times) and 1, giving

$$\tau(K_n) = \det \tilde{L} = n^{n-2},$$

recovering 9.5.

Definition 10.12 (Totally unimodular). A matrix is *totally unimodular* if every minor lies in $\{-1, 0, 1\}$.

Proposition 10.13. The oriented incidence matrix B of any graph is totally unimodular.

Proof. Induct on k ; the case $k = 1$ is immediate. Let M be a $k \times k$ submatrix of B on rows $V' \subseteq V$ and columns $S \subseteq E$, so each column has at most one $+1$ and at most one -1 . Then zero column $\Rightarrow \det M = 0$; every column has both signs $\Rightarrow \sum_{i \in V'} M_{i,\bullet} = 0 \Rightarrow \det M = 0$; some column j has a unique nonzero entry ± 1 in row $i \Rightarrow \det M = \pm \det M'$, with M' the $(k-1) \times (k-1)$ minor; $\det M' \in \{-1, 0, 1\}$ by induction. $\square$

Theorem 10.14. Say that the Laplacian eigenvalues of G are $\lambda_1, \dots, \lambda_{n-1}, \lambda_n = 0$, then

$$\tau(G) = \frac{1}{n} \prod_{i=1}^{n-1} \lambda_i.$$

Proof. Every first cofactor of L equals $\tau(G)$ by 10.10, so the sum of principal $(n-1) \times (n-1)$ minors of L is $n\tau(G)$. The λ -coefficient of $\det(\lambda I - L)$ equals both $(-1)^{n-1}n\tau(G)$ and $(-1)^{n-1} \prod_{i=1}^{n-1} \lambda_i$. $\square$

Definition 10.15 (Spectrum). The *spectrum* $\text{Spec}(G)$ of a graph G on n vertices is the multiset $\{\alpha_1, \dots, \alpha_n\}$ of eigenvalues of its adjacency matrix $A(G)$. Since $L\mathbf{1} = 0$, the all-ones vector is a Laplacian eigenvector with eigenvalue 0.

Example 10.16 (d -regular form). For a d -regular graph G on n vertices, $L = dI_n - A$, so Laplacian eigenvalues are $\{d - \alpha : \alpha \in \text{Spec}(G)\}$ with $A\mathbf{1} = d\mathbf{1}$ matching $\alpha = d$ to $\lambda_n = 0$. Hence 10.14 gives

$$\tau(G) = \frac{1}{n} \prod_{\alpha \in \text{Spec}(G), \alpha \neq d} (d - \alpha).$$

10.1. Spanning Trees of the Hypercube.

Definition 10.17 (Cartesian product). For graphs $G = (V_1, E_1)$ and $H = (V_2, E_2)$, the *Cartesian product* $G \times H$ has vertex set $V_1 \times V_2$ and edges $\{(u, v), (u', v')\}$ with either $u = u'$ and $\{v, v'\} \in E_2$, or $\{u, u'\} \in E_1$ and $v = v'$.

Lemma 10.18. If $\text{Spec}(G) = \{\alpha_1, \dots, \alpha_m\}$ and $\text{Spec}(H) = \{\beta_1, \dots, \beta_n\}$ as multisets, then $\text{Spec}(G \times H) = \{\alpha_i + \beta_j : i \in [m], j \in [n]\}$.

Proof. $A(G \times H) = A(G) \otimes I_n + I_m \otimes A(H)$. For $A(G)u_i = \alpha_i u_i$ and $A(H)v_j = \beta_j v_j$,

$$A(G \times H)(u_i \otimes v_j) = (\alpha_i + \beta_j)(u_i \otimes v_j),$$

and the mn tensors $u_i \otimes v_j$ form an orthogonal basis of $\mathbb{R}^{mn}$. $\square$

Definition 10.19 (Hypercube). H_d is the 1-skeleton of the d -dimensional cube: $V(H_d) = \{0, 1\}^d$ with edges joining vectors at Hamming distance 1. Equivalently $H_d = K_2 \times K_2 \times \dots \times K_2$ (d factors); it has 2^d vertices and is d -regular.

Theorem 10.20.

$$\tau(H_d) = \frac{1}{2^d} \prod_{k=1}^d (2k) \binom{d}{k} = 2^{2^d - d - 1} \prod_{k=1}^d k \binom{d}{k}.$$

Proof. $\text{Spec}(K_2) = \{+1, -1\}$, so iterating 10.18 gives $\text{Spec}(H_d) = \{\sum_{i=1}^d \varepsilon_i : \varepsilon \in \{\pm 1\}^d\}$, i.e. adjacency eigenvalue $d - 2k$ with multiplicity $\binom{d}{k}$ for $k = 0, \dots, d$. Hence Laplacian eigenvalue $2k$ with multiplicity $\binom{d}{k}$. Apply the d -regular form of 10.14 with $n = 2^d$ and Laplacian eigenvalues $2k$ of multiplicity $\binom{d}{k}$ for $k = 1, \dots, d$. The second equality uses $\sum_{k=0}^d \binom{d}{k} = 2^d$ and $\sum_{k=1}^d \binom{d}{k} = 2^d - 1$ to extract the powers of 2. $\square$

Example 10.21. $\tau(H_3) = 2^{2^3 - 3 - 1} \cdot 1 \binom{3}{1} \cdot 2 \binom{3}{2} \cdot 3 \binom{3}{3} = 2^4 \cdot 2^3 \cdot 3 = 3 \cdot 2^7 = 384$.

10.2. Reciprocity Formula.

Definition 10.22 (f_G). For a simple graph G on $V = [n]$, let G^+ be G with an extra vertex 0 joined to every vertex of G . Define

$$f_G(x) = \sum_T x^{\deg_T(0)-1},$$

summing spanning trees T of G^+ .

Definition 10.23 (F_G). For a simple graph G on $V = [n]$, with G^+ as in 10.22, define

$$F_G(x; x_1, \dots, x_n) = \sum_T x^{\deg_T(0)-1} \prod_{i=1}^n x_i^{\deg_T(i)-1},$$

summing over spanning trees T of G^+ .

Theorem 10.24. For $\bar{G}$ the complement of G on $[n]$ and $h = \sum_{i=1}^n x_i$,

$$F_{\bar{G}}(x, x_1, \dots, x_n) = (-1)^{n-1} F_G(-x - h, x_1, \dots, x_n).$$

Proof. Each tree T of G^+ has $\deg_T(0) + \sum_{i=1}^n \deg_T(i) = 2n$, hence $(\deg_T(0) - 1) + \sum_{i=1}^n (\deg_T(i) - 1) = n - 1$, so $F_G(x; x_1, \dots, x_n)$ is homogeneous of total degree $n - 1$ with $\deg(x) = \deg(x_i) = 1$. Since $\deg(h) = 1$, the substitution $x \mapsto -x - h$ preserves homogeneity, so

$$P(x; x_1, \dots, x_n) := F_{\bar{G}}(x, x_1, \dots, x_n) - (-1)^{n-1} F_G(-x - h, x_1, \dots, x_n)$$

is homogeneous of total degree $\leq n - 1$. Setting $x_i = 0$ kills trees with $\deg_T(i) \geq 2$; the surviving trees attach i as a leaf to a unique neighbor $j \in V(G^+) \setminus \{i\}$, and removing i identifies them with spanning trees of $(G - i)^+$ on $n - 1$ vertices. The induction hypothesis on $n - 1$ then gives $P|_{x_i=0} = 0$. Hence $x_1 \cdots x_n \mid P$; the divisor has total degree n , exceeding $\deg P \leq n - 1$, so $P = 0$. Base case $n = 1$ is direct. $\square$

Theorem 10.25 (Reciprocity Formula). $f_{\bar{G}}(x) = (-1)^{n-1} f_G(-x - n)$.

Proof. Set $x_1 = \cdots = x_n = 1$ in 10.24. $\square$

Lemma 10.26. $f_G(0) = n \tau(G)$, and $[x^{n-1}] f_G = 1$.

Proof. $f_G(0)$ extracts trees with $\deg_T(0) = 1$: a single edge $\{0, i\}$ together with a spanning tree of G , giving $n \tau(G)$ summands. The leading term comes from the unique tree with $\deg_T(0) = n$. $\square$

Remark 10.27. $f_{\bullet}(x) = 1$, so by iterated 10.28 $f_{\bar{K}_n}(x) = x^{n-1}$. Then 10.25 gives

$$f_{K_n}(x) = (-1)^{n-1} (-x - n)^{n-1} = (x + n)^{n-1},$$

and 10.26 yields $\tau(K_n) = n^{n-1}/n = n^{n-2}$, recovering 9.5.

Lemma 10.28. For disjoint graphs G and H , $f_{G \sqcup H}(x) = x f_G(x) f_H(x)$.

Proof. A spanning tree of $(G \sqcup H)^+$ is determined by its restrictions $T_G \subseteq G^+$ and $T_H \subseteq H^+$ (each a spanning tree on its side) and contributes $x^{\deg_{T_G}(0) + \deg_{T_H}(0) - 1} = x \cdot x^{\deg_{T_G}(0) - 1} x^{\deg_{T_H}(0) - 1}$. $\square$

Example 10.29 (Complete bipartite graph). $\overline{K_m \sqcup K_n} = K_{m,n}$ on $m + n$ vertices. By 10.28, $f_{K_m \sqcup K_n}(x) = x(x + m)^{m-1} (x + n)^{n-1}$. Applying 10.25,

$$f_{K_{m,n}}(x) = (x + m + n)(x + n)^{m-1} (x + m)^{n-1},$$

and 10.26 gives $\tau(K_{m,n}) = n^{m-1} m^{n-1}$.

10.3. Weighted Matrix–Tree Theorem.

Definition 10.30 (Weighted Laplacian). For $G = (V, E)$ on $V = [n]$ with edge weights $x_e \in \mathbb{R}_{\geq 0}$, the *weighted Laplacian* $L_w = (L_{ij})$ is

$$L_{ij} = \begin{cases} -\sum_{e=ij} x_e & i \neq j, \\ \sum_{e \ni i} x_e & i = j. \end{cases}$$

For a spanning tree T of G , $\text{wt}(T) = \prod_{e \in T} x_e$.

Theorem 10.31 (Weighted Matrix–Tree Theorem). $\sum_T \text{wt}(T) = \det \widetilde{L}_w$, *summing over spanning trees T of G , for any choice of deleted row/column.*

Proof. Both sides are polynomials in $\{x_e\}_{e \in E}$. For $x_e \in \mathbb{Z}_{\geq 0}$, replace each edge e by x_e parallel copies to obtain a multigraph $\widetilde{G}$; spanning trees of $\widetilde{G}$ are pairs (spanning tree $T \subseteq E(G)$, choice of one parallel copy per $e \in T$), so $\tau(\widetilde{G}) = \sum_T \text{wt}(T)$. The unweighted Laplacian of $\widetilde{G}$ equals L_w entrywise, so 10.10 applies. Polynomial agreement on $\mathbb{Z}_{\geq 0}^E$ forces equality on $\mathbb{R}_{\geq 0}^E$. $\square$

10.4. Directed Matrix–Tree Theorem.

Definition 10.32 (Digraph Laplacian). For a digraph $G = (V, E)$ on $V = [n]$, let the adjacency matrix $A = (a_{ij})$ with $a_{ij} = |\{e \in E : e = (i, j)\}|$, and $D = \text{diag}(d_1^{\text{in}}, \dots, d_n^{\text{in}})$ with $d_j^{\text{in}} = \text{indeg}_G(j)$. The *(in-)Laplacian* is $L = D - A$; every column of L sums to zero.

Definition 10.33 (Arborescence). An *arborescence* of G rooted at $r \in V$ is a spanning subgraph T that is a tree as undirected graph and whose edges are all directed away from r , equivalently, every $v \in V$ admits a unique directed path from r to v in T . Write $\text{Arb}_r(G)$ for the number of arborescences of G rooted at r .

Lemma 10.34. *For $e : i \rightarrow j$ with $j \neq r$, let $G_1 = G - e$ and $G_2 = G$ minus all edges into j except e . Then*

$$\text{Arb}_r(G) = \text{Arb}_r(G_1) + \text{Arb}_r(G_2).$$

Proof. Each arborescence T has $\text{indeg}_T(j) = 1$. Splitting on whether the unique edge of T entering j equals e partitions arborescences of G into those of G_2 (when it does) and those of G_1 (when it does not). $\square$

Lemma 10.35. *Fix $r \in [n]$. Then L^{kr} is independent of $k \in [n]$.*

Proof. Every column of L sums to zero, so any column is minus the sum of the others. The minors $\det(L \setminus \text{row } k, \text{col } r)$ therefore agree up to sign as k varies, and

$$L^{kr} = (-1)^{k+r} \det(L \setminus \text{row } k, \text{col } r)$$

has the same value for every k . $\square$

Theorem 10.36 (Directed Matrix–Tree Theorem). *For all $r, k \in [n]$,*

$$\text{Arb}_r(G) = L^{kr}.$$

Proof. Edges into r change neither side; assume there are none. Induct on $|E|$.

Reduction step: with notation of 10.34, L_G , L_{G_1} , L_{G_2} agree outside column j , and $(L_G)_{\bullet, j} = (L_{G_1})_{\bullet, j} + (L_{G_2})_{\bullet, j}$. Multilinearity in column j gives $L_G^{kr} =$

$L_{G_1}^{kr} + L_{G_2}^{kr}$, matching 10.34 by the inductive hypothesis whenever $|E(G_2)| < |E|$, i.e. $\text{indeg}_G(j) \geq 2$.

Base: $\text{indeg}_G(j) = 1$ for every $j \neq r$.

- G is an arborescence rooted at r : order vertices so $r = 1$ and edges go to higher indices; L is lower triangular with diagonal $(0, 1, \dots, 1)$, hence $L^{kr} = 1 = \text{Arb}_r(G)$.
- G contains a directed cycle (necessarily disjoint from r): the columns of L indexed by that cycle sum to zero, so $L^{kr} = 0 = \text{Arb}_r(G)$.

Independence of k follows from 10.35. $\square$

Example 10.37 (Increasing trees). For the tournament G on $[n]$ with edges $i \rightarrow j$ iff $i < j$, $L^{11} = (n-1)!$, and arborescences rooted at 1 are exactly the increasing trees on $[n]$. Hence the number of increasing trees on $[n]$ is $(n-1)!$.

Theorem 10.38 (Abel's Binomial Formula). *For variables x, y, z and $n \geq 0$,*

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} y (y+kz)^{k-1} (x-kz)^{n-k}.$$

Proof. By homogeneity in (x, y, z) , divide by z^n and replace (x, y) by $(x/z, y/z)$ to reduce to $z = 1$, i.e.

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} y (y+k)^{k-1} (x-k)^{n-k}.$$

Let $V = \{A, B, 1, \dots, n\}$ with weights $a_A = y$, $a_B = x - n$, $a_i = 1$ ($i \in [n]$). By 9.3,

$$\sum_T \prod_{v \in V} a_v^{\deg_T(v)-1} = \left(\sum_{v \in V} a_v \right)^n = (y + (x-n) + n)^n = (x+y)^n.$$

For a tree T on V , let $A-C$ be the first edge on the unique $A-B$ path, and delete it. Writing the resulting components as $T_A \ni A$ and $T_B \ni B$ (with $C \in V(T_B)$),

$$\prod_{v \in V} a_v^{\deg_T(v)-1} = a_A a_C \left(\prod_{v \in V(T_A)} a_v^{\deg_{T_A}(v)-1} \right) \left(\prod_{v \in V(T_B)} a_v^{\deg_{T_B}(v)-1} \right).$$

Set $S = V(T_A) \cap [n]$ and $|S| = k$ (so $\binom{n}{k}$ choices). By 9.3 on $\{A\} \cup S$ and on $\{B\} \cup S^c$,

$$\begin{aligned} \sum_{T_A} \prod_v a_v^{\deg_{T_A}(v)-1} &= (a_A + k)^{k-1} = (y+k)^{k-1}, \\ \sum_{T_B} \prod_v a_v^{\deg_{T_B}(v)-1} &= (a_B + n - k)^{n-k-1}. \end{aligned}$$

Moreover $a_A = y$, and $\sum_{C \in V(T_B)} a_C = a_B + (n-k) = x-k$, hence the B -side contributes $(x-k)^{n-k}$. Since $T \mapsto (S, T_A, T_B, C)$ is a bijection,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} y (y+k)^{k-1} (x-k)^{n-k}.$$

Undoing the reduction gives the general z case. $\square$

Remark 10.39 (Weighted Directed Matrix-Tree Theorem). The same proof, applied to a weighted digraph Laplacian, gives $\sum_T \prod_{e \in T} x_e = L^{kr}$, summing over arborescences rooted at r .

10.5. Electrical Networks.

Definition 10.40 (Electrical resistor network). $G = (V, E)$ on $V = [n]$ with edge weights $x_e = 1/R_e > 0$ (*conductances*); $R_e > 0$ is the *resistance* of e . Fix an orientation on E ; on each directed edge $e = u \rightarrow v$ assign a current $I_e \in \mathbb{R}$ and a voltage $V_e \in \mathbb{R}$.

Definition 10.41 (First Kirchhoff's Law). At every $v \in V$,

$$\sum_{e: * \rightarrow v} I_e = \sum_{e: v \rightarrow *} I_e.$$

Definition 10.42 (Second Kirchhoff's Law). For every cycle $C = (e_1, \dots, e_m)$ in G , $\sum_{i=1}^m \varepsilon_i V_{e_i} = 0$, with $\varepsilon_i \in \{\pm 1\}$ recording e_i 's orientation against the cycle traversal.

Lemma 10.43. *Kirchhoff's voltage law holds iff there exists $U : V \rightarrow \mathbb{R}$ with $V_e = U_v - U_u$ for every directed edge $e = u \rightarrow v$.*

Proof. If $V_e = U_v - U_u$ on $e = u \rightarrow v$, then along any cycle $C = (e_1, \dots, e_m)$ the signed sum $\sum_{i=1}^m \varepsilon_i V_{e_i}$ telescopes to 0. Converse is true by path independence. $\square$

Definition 10.44 (Ohm's law). For every edge e , $V_e = I_e R_e$.

Definition 10.45 (Kirchhoff matrix). $K = (K_{ij}) \in \mathbb{R}^{n \times n}$ with

$$K_{ij} = \begin{cases} -\sum_{e: e=ij} 1/R_e, & i \neq j, \\ \sum_{e: i \in e} 1/R_e, & i = j. \end{cases}$$

Equivalently, $K = L_w$ in the sense of 10.30 with $x_e = 1/R_e$.

Proposition 10.46 (Kirchhoff system). *Inject external current I_{tot} at vertex 1 and extract it at vertex n . Under 10.43 and Ohm's law, the potential vector $\vec{U} = (U_1, \dots, U_n)^T$ satisfies*

$$K\vec{U} = (I_{\text{tot}}, 0, \dots, 0, -I_{\text{tot}})^T.$$

Proof. For $v \notin \{1, n\}$ with neighbors $s_1, \dots, s_d$ along edges of resistance $R_1, \dots, R_d$, Ohm's law gives $I_{e_i} = (U_v - U_{s_i})/R_i$ on the edge oriented $s_i \rightarrow v$ and Kirchhoff's current law forces

$$\left(\sum_{i=1}^d 1/R_i\right)U_v - \sum_{i=1}^d (1/R_i)U_{s_i} = 0,$$

i.e. $(K\vec{U})_v = 0$. At $v = 1$ and $v = n$ the same identity holds with right-hand side $\pm I_{\text{tot}}$. $\square$

Definition 10.47 (Effective resistance). Fix $U_n = 0$. The *effective resistance* between 1 and n is

$$R_{\text{eff}} = \frac{U_1 - U_n}{I_{\text{tot}}} = \frac{U_1}{I_{\text{tot}}}.$$

Theorem 10.48. *Let $\tilde{K}$ denote K with row and column n deleted, and $\tilde{\tilde{K}}$ denote K with rows and columns 1 and n deleted. Then*

$$R_{\text{eff}} = \frac{\det \tilde{\tilde{K}}}{\det \tilde{K}}.$$

Proof. Setting $U_n = 0$, 10.46 restricts to

$$\tilde{K}(U_1, \dots, U_{n-1})^\top = (I_{\text{tot}}, 0, \dots, 0)^\top.$$

By Cramer's Rule on the first coordinate, $U_1 = I_{\text{tot}} \det \tilde{\tilde{K}} / \det \tilde{K}$, so $R_{\text{eff}} = U_1 / I_{\text{tot}} = \det \tilde{\tilde{K}} / \det \tilde{K}$. $\square$

Theorem 10.49. *Let $\tilde{G}$ be the multigraph obtained from G by identifying vertices 1 and n , with edge weights $1/R_e$. Then*

$$R_{\text{eff}} = \frac{\sum_{T' \text{ sp. tree of } \tilde{G}} \text{wt}(T')}{\sum_{T \text{ sp. tree of } G} \text{wt}(T)}.$$

Proof. By 10.31, $\det \tilde{K} = \sum_T \text{wt}(T)$ over spanning trees of G . The Laplacian of $\tilde{G}$ has the merged vertex's row/column equal to the sum of rows/columns 1 and n of K ; deleting that row/column equals deleting both rows/columns 1 and n from K , hence equals $\tilde{\tilde{K}}$. Apply 10.31 again to obtain $\det \tilde{\tilde{K}} = \sum_{T'} \text{wt}(T')$ over spanning trees of $\tilde{G}$. $\square$

Example 10.50 (Series connection). G : path 1–2–3 with resistances R_1, R_2 ; the unique spanning tree has $\text{wt} = 1/(R_1 R_2)$. $\tilde{G}$: two parallel edges between two vertices, with two spanning trees of weights $1/R_1$ and $1/R_2$. Hence

$$R_{\text{eff}} = \frac{1/R_1 + 1/R_2}{1/(R_1 R_2)} = R_1 + R_2.$$

Example 10.51 (Parallel connection). G : two parallel edges with conductances $1/R_1, 1/R_2$ between two vertices. Then

$$R_{\text{eff}}^{-1} = \frac{1}{R_1} + \frac{1}{R_2}.$$

Remark 10.52 (Y – Δ transform). Any star (Y) on three leaves with resistances R_1, R_2, R_3 admits unique triangle (Δ) resistances R'_1, R'_2, R'_3 such that effective resistances between every pair of leaves agree; explicitly $R'_i = (R_1 R_2 + R_2 R_3 + R_3 R_1) / R_i$, with R'_i the triangle edge opposite the leaf incident to R_i .

10.6. Random Walks on Graphs.

Definition 10.53 (Simple random walk on a path). On $\{0, 1, \dots, n\}$, from $0 < i < n$ move to $i \pm 1$ each with probability $1/2$; absorb at 0 and n . Let $p(i)$ be the probability of reaching n before 0 starting at i .

Proposition 10.54. $p(i) = i/n$ for $0 \leq i \leq n$.

Proof. Conditioning on the first step gives $p(i) = \frac{1}{2}p(i+1) + \frac{1}{2}p(i-1)$ for $0 < i < n$, with $p(0) = 0$ and $p(n) = 1$. The unique solution of this two-point boundary-value problem on $\{0, \dots, n\}$ is linear, hence $p(i) = i/n$. $\square$

Definition 10.55 (Random walk on a weighted graph). $G = (V, E)$ on $V = [n]$ with edge weights $x_e > 0$ and distinguished vertices $A, B \in V$. From $v \notin \{A, B\}$ with incident edges $e_1, \dots, e_d$ to neighbors $\tilde{v}_1, \dots, \tilde{v}_d$, transition to $\tilde{v}_i$ with probability $x_{e_i} / \sum_{j=1}^d x_{e_j}$; the walk absorbs at A and B . Set

$$p(v) = \mathbb{P}(\text{walk from } v \text{ hits } A \text{ before } B).$$

Theorem 10.56. *With $A = 1$, $B = n$, $p(1) = 1$, $p(n) = 0$, and at every $v \notin \{1, n\}$,*

$$\left(\sum_{i=1}^d x_{e_i}\right)p(v) - \sum_{i=1}^d x_{e_i} p(\tilde{v}_i) = 0,$$

i.e. $L_w \vec{p} = (, 0, \dots, 0, -*)^\top$ for the weighted Laplacian with x_e .*

Corollary 10.57. *For the network on G with conductances $x_e = 1/R_e$ and boundary potentials $U_1 = 1$, $U_n = 0$,*

$$p(v) = U_v \quad \forall v \in V.$$

Proof. By 10.56, $\vec{p}$ solves $K\vec{p} = (*, 0, \dots, 0, -*)^\top$ for $v \notin \{1, n\}$ with $p(1) = 1$, $p(n) = 0$. Rescaling I_{tot} in 10.46 so that $U_1 = 1$, $\vec{U}$ solves the same problem. $\tilde{K}$ is invertible by 10.10, so $p(v) = U_v$ for all v . $\square$

11. PARKING FUNCTIONS

Definition 11.1 (Preference function). Fix n cars and n parking spots on a one-way road, both indexed by $[n]$. A *preference function* is a function

$$f : [n] \rightarrow [n], \quad i \mapsto f_i,$$

where f_i is the preferred spot of car i .

Definition 11.2 (Parking procedure). Cars arrive in the order $1, 2, \dots, n$. Car i first drives to spot f_i and parks there if it is empty; otherwise, it continues forward and parks in the first available spot, if one exists.

Definition 11.3 (Parking function). A preference function $f = (f_1, \dots, f_n)$ is a *parking function* if all n cars park under the parking procedure.

Lemma 11.4. *For $f = (f_1, \dots, f_n)$ with $f_i \in [n]$, the following are equivalent.*

- (i) *f is a parking function.*
- (ii) *For every $k = 1, \dots, n$,*

$$|\{i : f_i \geq n - k + 1\}| \leq k.$$

- (iii) *If $a_1 \leq \dots \leq a_n$ is the increasing rearrangement of $f_1, \dots, f_n$, then $a_i \leq i$ for all $i \in [n]$.*

- (iv) *There exists a permutation $w_1, \dots, w_n$ of $1, 2, \dots, n$ such that $f_i \leq w_i$ for all $i \in [n]$.*

Proof. Condition (ii) says that at most one car prefers spot n , at most two cars prefer spots in $\{n-1, n\}$, and so on. This is the same as (iii) after sorting. By assigning car i a distinct upper bound $w_i \in [n]$ with $w_i \geq f_i$ we can get condition (iv). The parking procedure succeeds exactly when no terminal block of k spots is claimed by more than k cars before earlier spots are considered. $\square$

Theorem 11.5. *The number of parking functions of size n is*

$$(n+1)^{n-1}.$$

Proof. Consider the modified parking procedure with n cars and $n+1$ parking spots on a circular road, oriented counterclockwise. Now every preference function $f : [n] \rightarrow [n+1]$ parks all cars and leaves exactly one spot empty. Let P_i be the

set of preference functions for which spot i is the empty spot. Cyclically shifting all preferences gives bijections $P_i \cong P_{i+1}$, so

$$|P_1| = |P_2| = \cdots = |P_{n+1}|.$$

Since there are $(n+1)^n$ preference functions total,

$$|P_{n+1}| = \frac{(n+1)^n}{n+1} = (n+1)^{n-1}.$$

Finally, P_{n+1} is exactly the set of ordinary parking functions on the one-way road with spots $[n]$: a car reaches spot $n+1$ in the circular model precisely when the corresponding one-way procedure would fail. $\square$

11.1. Labeled Dyck Paths.

Definition 11.6 (Labeled Dyck path). A *labeled Dyck path* of size n is a Dyck path of length $2n$ whose n up-steps are labeled by $1, 2, \dots, n$ without repetitions, such that labels increase along consecutive up-steps.

Theorem 11.7. *The number of labeled Dyck paths of size n is*

$$(n+1)^{n-1}.$$

Proof. There is a bijection from labeled Dyck paths to parking functions. Given a labeled Dyck path P , define $f = (f_1, \dots, f_n)$ by

$$f_i = j \quad \text{if the up-step labeled } i \text{ lies on the } j\text{th diagonal.}$$

The Dyck condition is exactly the parking-function inequality in 11.4, while the increasing-label condition ensures that this construction is reversible. Thus labeled Dyck paths of size n are counted by 11.5. $\square$

12. THE BEST THEOREM

Definition 12.1 (Eulerian cycle). Let $G = (V, E)$ be a finite directed graph. An *Eulerian cycle* is an ordering

$$C = (e_1, e_2, \dots, e_m)$$

of all edges in E , without repetitions, such that the endpoint of e_i is the beginning of e_{i+1} for every i , with $e_{m+1} = e_1$.

Theorem 12.2 (Euler's Criterion). *A finite directed graph $G = (V, E)$ has an Eulerian cycle if and only if*

- (a) G is connected as an undirected graph;
- (b) $\text{indeg}_G(v) = \text{outdeg}_G(v)$ for every $v \in V$.

Proof. Necessity is immediate from traversing an Eulerian cycle: every time the cycle enters a vertex, it also leaves it, and all edges lie in one undirected component. Conversely, under (a) and (b), start walking from any edge and keep taking unused outgoing edges until returning to the starting vertex. If unused edges remain, connectedness supplies a vertex on the current closed tour incident to an unused edge; splice in another closed tour. Induct on the number of unused edges. $\square$

Lemma 12.3. *If G is connected and $\text{indeg}_G(v) = \text{outdeg}_G(v)$ for every $v \in V$, then*

$$\text{Arb}_v(G) = \text{Arb}_{v'}(G) \quad \forall v, v' \in V.$$

Equivalently, all cofactors of the digraph Laplacian are equal.

Proof. Balancedness gives $\text{indeg}_G(v) = \text{outdeg}_G(v)$, so every row of the digraph Laplacian L also sums to zero. By 10.36, $\text{Arb}_r(G) = L^{kr}$; 10.35 makes L^{kr} independent of k , and the same cofactor argument applied to $L^\top$ (or to rows of L) makes L^{kr} independent of r . Hence $\text{Arb}_v(G) = \text{Arb}_{v'}(G)$ for all $v, v' \in V$. $\square$

Lemma 12.4 (Fixed-edge BEST bijection). *Fix an edge $e \in E$ and let r be the endpoint of e . Eulerian cycles beginning with e are in bijection with pairs*

$$(T, (\pi_v)_{v \in V}),$$

where T is an arborescence rooted at r , and π_v is a permutation of the set of incoming edges at v after deleting the unique edge of T entering v if $v \neq r$, and after deleting e if $v = r$.

Proof. Let $C = (e_1, \dots, e_m)$ with $e_1 = e$. For $v \neq r$, let f_v be the first edge of C entering v , and set $T := \{f_v : v \neq r\}$. Then each $v \neq r$ has exactly one incoming edge in T . If T contained a directed cycle, let u be a vertex on it whose first visit along C is minimal among such vertices; the marked edge into u on the cycle then comes from a vertex of the cycle visited strictly earlier, a contradiction. Hence T is an arborescence rooted at r .

Let π_v list the remaining incoming edges at v (omit f_v for $v \neq r$, omit e for $v = r$) in their order along C .

To recover C , reverse edges to G^{op} and traverse

$$e_1^{\text{op}}, e_m^{\text{op}}, e_{m-1}^{\text{op}}, \dots, e_2^{\text{op}}.$$

For $v \neq r$, f_v^{op} is the last exit from v in this tour. The last-exit rule (use each prescribed non-tree out-edge before the tree out-edge) reconstructs the unique Eulerian tour in G^{op} with that last-exit tree and these local orders; reversing again yields the unique Eulerian C in G with first edge e . $\square$

Theorem 12.5 (BEST Theorem). *Let $G = (V, E)$ be a connected directed graph with $\text{indeg}_G(v) = \text{outdeg}_G(v)$ for all $v \in V$. Then the number of Eulerian cycles in G is*

$$|E| \text{Arb}_r(G) \prod_{v \in V} (\text{indeg}_G(v) - 1)!,$$

where r is any vertex. Here $\text{Arb}_r(G)$ is the number of arborescences rooted at r .

Proof. By 12.3, the number $\text{Arb}_r(G)$ is independent of the root r . Fix an edge $e \in E$ and let r be the endpoint of e . By 12.4, the number of Eulerian cycles beginning with e is

$$\text{Arb}_r(G) \prod_{v \in V} (\text{indeg}_G(v) - 1)!$$

since the deleted edge leaves $\text{indeg}_G(v) - 1$ incoming edges to order at each vertex v . Multiplying by the $|E|$ choices of first edge gives the formula. $\square$

13. LINDSTRÖM'S LEMMA

Definition 13.1 (Sources and sinks). Let $G = (V, E)$ be an acyclic directed graph drawn in the plane with edge weights x_e for $e \in E$. Fix *sources* $A_1, \dots, A_n$ on the left boundary of G (listed top to bottom) and *sinks* $B_1, \dots, B_n$ on the right boundary.

Definition 13.2 (Non-crossing path family). An n -tuple $(P_1, \dots, P_n)$ of directed paths with $P_i : A_i \rightarrow B_i$ is *non-crossing* if the paths have no common vertex. For such a family,

$$N(A_1, \dots, A_n, B_1, \dots, B_n) = \sum_{(P_1, \dots, P_n)} \prod_{i=1}^n \text{wt}(P_i),$$

where $\text{wt}(P) = \prod_{e \in P} x_e$ and the sum is over non-crossing families.

Definition 13.3 (Path matrix). The *path matrix* is $C = (c_{ij})$ with

$$c_{ij} = \sum_{P: A_i \rightarrow B_j} \text{wt}(P),$$

summing over all directed paths from A_i to B_j (not necessarily non-crossing).

Theorem 13.4 (Lindström's Lemma, Gessel–Viennot Method).

$$N(A_1, \dots, A_n, B_1, \dots, B_n) = \det(C).$$

Proof. Expanding $\det(C)$,

$$\det(C) = \sum_{w \in S_n} (-1)^{\text{sgn}(w)} \prod_{i=1}^n c_{i, w_i} = \sum_{w \in S_n} \sum_{(P_1, \dots, P_n)} (-1)^{\text{sgn}(w)} \prod_{i=1}^n \text{wt}(P_i),$$

where $P_i : A_i \rightarrow B_{w_i}$. Non-crossing families contribute with $\text{sgn}(w) = \text{sgn}(\text{id}) = 1$. It suffices to show that every *bad* family, one in which some pair P_i, P_j shares a vertex, contributes net zero.

Take the minimal i such that P_i meets another path, then the first vertex C of P_i on another path, then the minimal $j \neq i$ with $C \in P_j$, and *tail-swap* at C : replace P_i, P_j by $\tilde{P}_i : A_i \rightarrow B_{w_j}$ and $\tilde{P}_j : A_j \rightarrow B_{w_i}$ obtained by following P_i to C , then P_j after C (and symmetrically for $\tilde{P}_j$). This involution reverses $\text{sgn}(w)$ and preserves total weight, pairing bad contributions. $\square$

13.1. Connected Skew Young Diagrams.

Definition 13.5 (Connected Skew Young diagram). For partitions $\lambda \supset \mu$, the *skew Young diagram* λ/μ is the set difference of the corresponding Young diagrams. It is *connected* if it is connected as a subgraph of the integer lattice.

Theorem 13.6. $N(n, k)$ equals the number of connected skew Young diagrams with k rows and $n + 1 - k$ columns. Equivalently, on the staircase grid it equals

$$\det \begin{pmatrix} \binom{n-1}{k-1} & \binom{n-1}{k-2} \\ \binom{n-1}{k} & \binom{n-1}{k-1} \end{pmatrix} = \binom{n-1}{k} \binom{n-1}{k-1} \frac{1}{k},$$

matching 2.33.

Proof. Biject connected skew diagrams with k rows to non-crossing path pairs $P_1 : A_1 \rightarrow B_1, P_2 : A_2 \rightarrow B_2$ on the staircase; apply 13.4. $\square$

Theorem 13.7. $N(n, k)$ equals the number of Dyck paths of length $2n$ with exactly k up-steps in odd positions $1, 3, 5, \dots$

Proof. See [1] for a proof of this theorem and a bijection to pairs of non-crossing free Dyck paths. $\square$

Definition 13.8 (Motzkin path). A *Motzkin path* of length n is a lattice path from $(0, 0)$ to $(n, 0)$ using up-, down-, and horizontal steps that never goes below the x -axis.

Remark 13.9. See [2] for the connection to Narayana numbers and connected skew Young diagrams.

14. PLANE PARTITIONS

Definition 14.1 (Plane partition). Fix $m, n, k \geq 1$. A *plane partition* of shape (m, n, k) is an $m \times n$ array of integers in $\{0, 1, \dots, k\}$ that weakly decreases along every row and every column. Write $PP(m, n, k)$ for the number of such arrays.

Proposition 14.2. $PP(m, n, 1) = \binom{m+n}{n}$.

Proof. Entries are 0 or 1 with weak decrease; the boundary between 1's and 0's is a lattice path from $(0, n)$ to $(m, 0)$. $\square$

Proposition 14.3.

$$PP(m, n, 2) = \det \begin{pmatrix} \binom{m+n}{m} & \binom{m+n}{m-1} \\ \binom{m+n}{m+1} & \binom{m+n}{m} \end{pmatrix} = N(m+n+1, m+1).$$

Proof. Plane partitions with entries in $\{0, 1, 2\}$ correspond to a pair (P_1, P_2) of weakly non-crossing lattice paths on the $m \times n$ grid, where P_1 separates entries ≥ 1 from 0 and P_2 separates entries 2 from ≤ 1 . Shift P_2 one step southeast to obtain a non-crossing pair with no common vertices; 13.4 gives the displayed 2×2 determinant. $\square$

Theorem 14.4.

$$PP(m, n, k) = \det(C), \quad C_{ij} = \binom{m+n}{m+i-j} \quad (1 \leq i, j \leq k).$$

Proof. Plane partitions with entries in $\{0, \dots, k\}$ correspond to k weakly non-crossing lattice paths $P_1, \dots, P_k$, where P_i separates entries $\geq k-i+1$ from entries $\leq k-i$. Shift P_i by $i-1$ steps southeast to obtain a non-crossing family with no common vertices; apply 13.4. $\square$

Theorem 14.5 (MacMahon).

$$PP(m, n, k) = \prod_{a=0}^{m-1} \prod_{b=0}^{n-1} \prod_{c=0}^{k-1} \frac{a+b+c+2}{a+b+c+1}.$$

Proof. See [11]. $\square$

14.1. Rhombus Tilings and Related Models.

Definition 14.6 (Hexagon and rhombus tiling). The hexagon $H_{m,n,k}$ has side lengths m, n, k . A *rhombus tiling* (or *lozenge tiling*) is a tiling of $H_{m,n,k}$ by congruent rhombi with angles 60° and 120° .

Theorem 14.7. $PP(m, n, k)$ equals the number of rhombus tilings of $H_{m,n,k}$.

Proof. A plane partition is a weakly decreasing $m \times n$ array with entries $0, 1, \dots, k$.

View it as a stack of unit cubes in an $m \times n \times k$ box—a three-dimensional Young diagram: place a_{ij} cubes above (i, j) in the xy -plane, with heights weakly decreasing in i and j .

Project the pile onto the three coordinate faces; each visible unit square corresponds to a rhombus in $H_{m,n,k}$, and the tiling is recovered from the stack. This gives a bijection between $PP(m, n, k)$ and rhombus tilings of $H_{m,n,k}$. $\square$

Definition 14.8 (Pseudoline arrangement). A *pseudoline arrangement* of type (m, n, k) is a collection of $m + n + k$ curves in the plane, each assigned to one of three types, with m , n , and k curves respectively, such that

- (i) curves of the same type do not intersect;
- (ii) curves of distinct types meet exactly once;
- (iii) no triple crossing.

Theorem 14.9. *Rhombus tilings of $H_{m,n,k}$ are in bijection with pseudoline arrangements of type (m, n, k) .*

Proof. From a rhombus tiling, the three tile orientations define $m + n + k$ monotone curves across the hexagon; extend them to pseudolines of the three types. From a pseudoline arrangement, read off the rhombus type at each elementary region between consecutive intersections; the local matching rules assemble a unique tiling of $H_{m,n,k}$. $\square$

Theorem 14.10. *$PP(m, n, k)$ equals the number of pseudoline arrangements of type (m, n, k) .*

Proof. By 14.4, a plane partition determines k weakly non-crossing lattice paths $P_1, \dots, P_k$ on the $m \times n$ grid, where P_i separates entries $\geq k - i + 1$ from entries $\leq k - i$. Treat the m , n , and k families of lines parallel to the three sides of $H_{m,n,k}$ as the three pseudoline types; the paths encode which tiles meet along each family, giving a pseudoline arrangement of type (m, n, k) . This is reversible. $\square$

Definition 14.11 (Honeycomb graph). $\text{Honey}_{m,n,k}$ is the honeycomb graph (dual of the triangular lattice) inside $H_{m,n,k}$.

Theorem 14.12.

$$PP(m, n, k) = \#\{\text{perfect matchings of } \text{Honey}_{m,n,k}\}.$$

Proof. Rhombus tilings of $H_{m,n,k}$ correspond to perfect matchings of $\text{Honey}_{m,n,k}$ by taking the dual of each rhombus tile. Apply 14.7. $\square$

Remark 14.13 (Arctic circle phenomenon). A uniformly random lozenge tiling of a large hexagon exhibits a frozen region near the boundary and a disordered region in the interior separated by a limiting circular arc (the arctic circle).

15. DELETION-CONTRACTION AND GRAPH POLYNOMIALS

Definition 15.1 (Deletion and contraction). For a graph $G = (V, E)$ and $e \in E$, the *deletion* $G \setminus e$ removes e ; the *contraction* G/e identifies the endpoints of e and deletes loops. A *subgraph* of G contains all vertices of V and a subset of E .

Definition 15.2 (Bridge and loop). An edge e is a *bridge* if $G \setminus e$ has more connected components than G ; a *loop* is an edge with both endpoints equal.

Definition 15.3. $F(G)$ is the number of forests; $C(G)$ is the number of connected spanning subgraphs.

Lemma 15.4. For every $e \in E$,

$$\tau(G) = \tau(G \setminus e) + \tau(G/e).$$

Proof. Partition spanning trees by whether e belongs to the tree. $\square$

15.1. Graph Colorings.

Definition 15.5 (k -coloring). A k -coloring of $G = (V, E)$ is a map $c : V \rightarrow \{1, \dots, k\}$ with $c(u) \neq c(v)$ for every edge $\{u, v\}$. The *chromatic number* $\chi(G)$ is the least k for which a k -coloring exists.

Theorem 15.6 (Chromatic Polynomial). *There exists a unique polynomial $\chi_G(t)$, the chromatic polynomial of G , such that $\chi_G(k)$ equals the number of k -colorings of G for every positive integer k .*

Proof. See [14]. $\square$

Example 15.7. $\chi_{K_3}(t) = t(t-1)(t-2)$ and $\chi_{P_3}(t) = t(t-1)^2$ for the triangle and the path on three vertices.

Lemma 15.8. For every non-loop edge $e = \{u, v\}$,

$$\chi_G(t) = \chi_{G \setminus e}(t) - \chi_{G/e}(t).$$

Proof. Induct on $|E|$. Base $|E| = 0$: $\chi_G(t) = t^{|V|}$. For $e = \{u, v\}$, colorings of $G \setminus e$ with $c(u) \neq c(v)$ are k -colorings of G ; those with $c(u) = c(v)$ biject to k -colorings of G/e . $\square$

15.2. Acyclic Orientations.

Definition 15.9 (Acyclic orientation). An *acyclic orientation* of G is an orientation of its edges with no directed cycle. Write $AO(G)$ for the number of acyclic orientations.

Example 15.10. $AO(C_n) = 2^n - 2$ for the n -cycle, and $AO(K_n) = n!$.

Theorem 15.11. For $n = |V|$,

$$AO(G) = (-1)^n \chi_G(-1).$$

Proof. See [12]. $\square$

Lemma 15.12. For every non-loop, non-bridge edge e ,

$$AO(G) = AO(G \setminus e) + AO(G/e).$$

Proof. By 15.8, $\tilde{\chi}_G(t) = (-1)^{|V|} \chi_G(t)$ satisfies $\tilde{\chi}_G(t) = \tilde{\chi}_{G \setminus e}(t) + \tilde{\chi}_{G/e}(t)$. Specializing to $t = -1$ gives the result. $\square$

15.3. The Tutte Polynomial.

Theorem 15.13 (Tutte Polynomial). *There exists a unique polynomial $T_G(x, y)$ named Tutte polynomial (or dichromatic polynomial) satisfying*

- (i) $T_G = 1$ if $E = \emptyset$;
- (ii) $T_G = T_{G \setminus e} + T_{G/e}$ if e is neither a loop nor a bridge;
- (iii) $T_G = x T_{G/e}$ if e is a bridge;
- (iv) $T_G = y T_{G \setminus e}$ if e is a loop.

Proof. See [13]. □

Proposition 15.14.

$$\begin{aligned} T_G(1, 1) &= \tau(G), \\ T_G(2, 1) &= F(G), \\ T_G(1, 2) &= C(G), \\ T_G(2, 0) &= AO(G). \end{aligned}$$

Moreover,

$$T_G(x, y) = \sum_{T \text{ spanning tree}} x^{|\text{int}(T)|} y^{|\text{ext}(T)|},$$

where $\text{int}(T)$ (resp. $\text{ext}(T)$) is the set of internally (resp. externally) active edges for a fixed order on E .

Proof. See [13]. □

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